IT318 : Web Technology Laboratory		
Teaching Scheme	**Examination Scheme**	
Lectures: 4 Hrs./Week	**Term Work:**	**NA**
	Oral :	**50 Marks**
	Practical:	**NA**
Credits: 2	**Total:**	**50 Marks**

Prerequisite Course: DBMSL

Course Objectives

1. To understand and apply the importance of website planning and website design issues.
2. To apply client side technologies for web application development.
3. To use server side technologies and database for web application development.

Course Outcomes (COs):

After successful completion of the course, student will be able to

Course Outcome (s)		Bloom's Taxonomy	
		Level	**Descriptor**
CO1	**Apply** the importance of website planning and website design issue.	3	Apply
CO2	**Apply** client side technologies for web application development.	3	Apply
CO3	**Use** server side technologies and database for web application development.	3	Apply

Mapping of Course Outcomes to Program Outcomes (POs) & Program Specific Outcomes (PSOs):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	3	1	2	2	2	2	1	2	3	3	-	-
CO2	2	2	3	2	3	2	2	2	2	1	2	3	3	-	1
CO3	2	2	3	2	3	2	2	2	2	1	2	3	3	-	1

Guidelines: This Web Technology Laboratory course has Web Technology as a core subject. The problem statements should be framed based on mentioned assignments in the syllabus for conduction of oral examination. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals, effective and efficient implementations in term work. Oral examination should be jointly conducted by the internal examiner and external examiner. All assignments are to be performed in Java Script, Servlet, JSP and PHP Language.

Term Work: Staff in-charge will suitably frame the above assignments and flexibility may be incorporated. Students will submit term work in the form of a journal. Each assignment has to be well documented with problem definition, code documented with comments. Staff in-charge will assess the assignments continuously and grade or mark each assignment on completion date. All the assignments should be conducted in Java Script, Servlet, JSP and PHP Language.

Suggested List of Assignments

Sr. No.	Assignments	No. of Hours	Cos

<table>
<tr><td>1</td><td>Case study: Before coding of the website, planning is important, students should visit different websites (Min. 5) for the different client projects and note down the evaluation results for these websites, either good website or bad website in following format.

<table><tr><td>Sr. No</td><td>Website URL</td><td>Purpose of Website</td><td>Things liked in the website</td><td>Things disliked in the website</td><td>Overall evaluation of the website (Good/Bad)</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>From the evaluation, students should learn and conclude different website design issues, which should be considered while developing a website.</td><td>4 Hrs.</td><td>CO1</td></tr>
<tr><td>2</td><td>Implement a web page index.html for any client website (e.g., a mobile shop website, Educational website or Clothes shop website) using following:
a. HTML syntax: heading tags, basic tags and attributes, frames, tables, images, lists, links for text and images, forms etc.
b. Use of Internal CSS, Inline CSS, External CSS</td><td>4 Hrs.</td><td>CO2</td></tr>
<tr><td>3</td><td>Implement an application in Java Script using following:
a) Design UI of application using HTML, CSS etc.
b) Include Java script validation
c) Use of prompt and alert window using Java Script
e.g., Design and implement a simple calculator using Java Script for operations like addition, multiplication, subtraction, division, square of number etc.
a) Design calculator interface like text field for input and output, buttons for numbers and operators etc.
b) Validate input values
c) Prompt/alerts for invalid values etc.</td><td>4 Hrs.</td><td>CO2</td></tr>
<tr><td>4</td><td>Design an application using Angular JS.
e.g., Design registration (first name, last name, username, and password) and login page using Angular JS.</td><td>4 Hrs.</td><td>CO2</td></tr>
<tr><td>5</td><td>Implement the sample program demonstrating the use of Servlet.
e.g., Create a database table ebookshop (book_id, book_title, book_author, book_price, quantity) using database like Oracle/MySQL etc. and display (use SQL select query) the table content using servlet.</td><td>4 Hrs.</td><td>CO3</td></tr>
<tr><td>6</td><td>Implement the program demonstrating the use of JSP.
e.g., Create a database table students_info (stud_id, stud_name, class, division, city) using database like Oracle/MySQL etc. and display (use SQL select query) the table content using JSP.</td><td>4 Hrs.</td><td>CO3</td></tr>
<tr><td>7</td><td>Build a dynamic web application using PHP and MySQL.
a. Create database tables in MySQL and create connection with PHP.
b. Create the add, update, delete and retrieve functions in the PHP web app interacting with MySQL database.</td><td>4 Hrs.</td><td>CO3</td></tr>
<tr><td>8</td><td>Design a login page with entries for name, mobile number email id and login button. Use struts and perform following validations
a. Validation for correct names
b. Validation for mobile numbers
c. Validation for email id
d. Validation if no entered any value
e. Re-display for wrongly entered values with message</td><td>4 Hrs.</td><td>CO3</td></tr>
</table>

	f. Congratulations and welcome page upon successful entries		
9	Design a website using Content management system of WordPress. Make the use of different plugins and themes of the WordPress.	4 Hrs.	CO3
10	Study of advanced tools: UI/UX(Figma), Laravel, Ruby on Rails, etc	4 Hrs.	CO3

Text Books:

1. Jeffrey C.Jackson, "Web Technologies: A Computer Science Perspective", Second Edition, Pearson Education, 2007, ISBN 978-0131856035.
2. Robert W. Sebesta, “Programming the World Wide Web”, 4th Edition, Pearson education, 2008..
3. Jason Hunter, “Java Servlet Programming”, 2nd Edition, O'reilly Publications.

Reference Books:

1. Roger S. Pressman, David Lowe, “Web Engineering”, Tata Mcgraw Hill Publication, 2007.
2. Kogent Learning Solution Inc., “Web Technologies” HTML, JavaScript, PHP, Java, JSP, XML and AJAX, Black Book, Dream Tech Press, 2014.
3. Jim Keogh, “J2EE: The Complete Reference”, 1st Edition , Tata McGraw Hill Pulishing Company, ISBN: 978-0-07-052912-0.
4. Marty Hall, Larry Brown,"Core Web Programming", Second Edition, Pearson Education, 2001, ISBN 978-0130897930.

Assignment No.1:
Case Study

Title:

Before coding of the website, planning is important, students should visit different websites (Min. 5) for the different client projects and note down the evaluation results for these websites, either good website or bad website in following format.

Sr. No	*Website URL*	*Purpose of Website*	*Things liked in the website*	*Things disliked in the website*	*Overall evaluation of the website (Good/Bad)*

From the evaluation, students should learn and conclude different website design issues, which should be considered while developing a website.

Theory:

Website: Website is a collection of related web pages that may contain text, images, audio, and video, etc. The first page of a website is called a home page. Each website has a specific internet address (URL) that you need to enter in your browser to access a website.

A website is hosted on one or more servers and can be accessed by visiting its homepage using a computer network. A website is managed by its owner that can be an individual, company, or organization.

The first website was introduced on 6 August 1991. It was developed by Tim Berners-Lee at CERN. As of January 2018, there were around 1.7 billion websites on the internet.

What is the need for a website?

A popular website is another name of success in modern times as today we are living in a digital world where people are spending most of their spare time on internet browsing news, images, videos, and more. A website is the first thing that appears on the screen of your laptop when you search for a product or service, or any other information online. So, it is very important to have a website whether you are a businessman, organization, online store, etc. Let us understand how a website can help you grow your business!

The benefits of having a website are as follows:

1. Cost-effective: The marketing of products and services through a website is less costly than print advertising, TV, and radio advertising. It also allows you to make changes related to your products and services such as you can edit products' specifications, features, discount offers, etc.

2. Broder demographic reach: A website can help your business reach people with diverse backgrounds throughout the world.

3. Business credibility: Having a website that is user-friendly and professional looking for your business makes people consider your business more credible than others who don't have a website to represent their businesses online.

4. 24/7 availability: You can keep your products available for sale around the clock through a website. Customers can buy anytime, and thus you can earn more profits than a traditional storefront.

5. Consumer convenience: It improves the shopping experience of your customers; they can buy your products from the comfort of their home.

6. Build customer rapport: Online presence helps you build and maintain a rapport among your customers. You can also introduce special offers or discounts for your loyal customers.

7. Improves productivity: It saves your time as you don't need to explain your products or services to each customer separately, as all the information is already available on the website.

8. Educate your customers: You can educate people about the benefits of your products and can show them how they are betters than the competitors' products.

9. Change location: You can move your business to another location without affecting your sales as customers still can find you through your website.

10. Grow manpower: You can post job openings for vacant positions and allow applicants to check job descriptions and apply online.

11: Personalized email address: It allows you to set up personalized email addresses for employees, companies, partners, etc. You can receive mails on these mail ids and reply accordingly. Furthermore, you can share offers, happy hours, updates, and can get feedback through email.

12. Improve customer service: It allows you to improve customer services such as you can share phone numbers to contact you, and display other relevant information such as order status, delivery status, refund or replacement procedure, etc.

Types of Websites:

A website can be of two types based on the type of webpages:

- Static Website
- Dynamic Website

Benefits of a Static Website:

- It takes less time to create a static website.
- It is less costly to develop this website.
- The webpages are delivered at high speed by the servers.
- Hosting through dedicated servers is available at a low price.
- It is more secure than dynamic sites as it does not rely on CMS plugins and server-side scripting.

Benefits of a Dynamic Website:

1. It is very quick and easy to make changes in a dynamic website as it is integrated with high-end programming.
2. The webpages are easier to maintain despite of being full of information and graphics.

3. E-commerce sites are dynamic as they need to be impressive and have easy navigation from one page to other pages.
4. It makes your site look professional.
5. The users can get customized information as per their interests, needs, and profile, etc. For example, facebook.com, flipkart.com, etc.

Difference between Static Website and Dynamic Website:

Static Website	Dynamic Website
As the name suggests, Its webpages do not change in terms of design, content, etc. The information or content remains the same.	As the name suggests, the webpages keep changing as pe users' requirements such as Facebook profile pages and an E commerce site. So, the content does not remain the same.
It mainly uses HTML and CSS and does not require server-side scripting, application server, and database.	It requires server-side scripting, application server, an database to create and send dynamic webpages to the client.
It has a limited number of pages.	It may contain thousands of pages in the database.
Its hosting cost is low, as HTML files need less space on the server.	Its hosting cost is higher as dynamic pages need more space o the server.
It requires low maintenance.	It requires high maintenance.
It loads quickly as it involves the use of mark-up languages to create a webpage.	It takes more time to load due to the more processing time.
It lacks the Content management feature.	It makes use of the Content Management Feature.
The content of the webpage cannot be changed during runtime.	The webpage content can be changed during runtime.
It does not require interaction with a database.	Interaction with the database occurs.
It is more secure or fewer chances of it getting hacked as it doesn't use plugins.	It is less secure and may get hacked easily as it uses man plugins and content sources.
It is more reliable, e.g., whenever the server is down, it is redirected to other nearby nodes.	It is less reliable, as it may go down for hours if the serve breaks down.

Website Design Issues:

1. Ensuring your website is fully responsive.
2. Successfully juggling form and function.
3. Designing for accessible experiences.
4. Making sure your site loads quickly.
5. Incorporating SEO best practices into site design.
6. Prioritizing user experience.
7. Creating an engaging website that converts.
8. Making a website that's intuitive to navigate.
9. Including visual elements without adding too much load time.
10. Ensuring your website looks great and performs well regardless of the browser.
11. Creating a functional, aesthetically pleasing site with a small team or limited resources.
12. Using the right amount of white space.
13. Including videos where helpful without making your page too slow.
14. Making visually appealing CTAs.
15. Maintaining security.

Conclusion: Thus Students have studied what is website? Its design, types of websites and its different design issues.

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Assignment No.2
HTML and CSS

Title: Implement a web page index.html for any client website (e.g., a mobile shop website, Educational website or Clothes shop website) using following:

a) HTML syntax: heading tags, basic tags and attributes, frames, tables, images, lists, links for text and images, forms etc.
b) Use of Internal CSS, Inline CSS, External CSS.

Theory:

What is HTML?

HTML is an acronym which stands for **Hyper Text Markup Language** which is used for creating web pages and web applications. Let's see what is meant by Hypertext Markup Language, and Web page.

Hyper Text: HyperText simply means "Text within Text." A text has a link within it, is a hypertext. Whenever you click on a link which brings you to a new webpage, you have clicked on a hypertext. HyperText is a way to link two or more web pages (HTML documents) with each other.

Markup language: A markup language is a computer language that is used to apply layout and formatting conventions to a text document. Markup language makes text more interactive and dynamic. It can turn text into images, tables, links, etc.

Web Page: A web page is a document which is commonly written in HTML and translated by a web browser. A web page can be identified by entering an URL. A Web page can be of the static or dynamic type. **With the help of HTML only, we can create static web pages**.

Hence, HTML is a markup language which is used for creating attractive web pages with the help of styling, and which looks in a nice format on a web browser. An HTML document is made of many HTML tags and each HTML tag contains different content.

Let's see a simple example of HTML.

```
<!DOCTYPE>
<html>
<head>
<title>Web page title</title>
</head>
<body>
<h1>Write Your First Heading</h1>
<p>Write Your First Paragraph.</p>
</body>
</html>
```

Description of HTML Example

<!DOCTYPE>: It defines the document type or it instruct the browser about the version of HTML.

<html > :This tag informs the browser that it is an HTML document. Text between html tag describes the web document. It is a container for all other elements of HTML except <!DOCTYPE>

<head>: It should be the first element inside the <html> element, which contains the metadata(information about the document). It must be closed before the body tag opens.

<title>: As its name suggested, it is used to add title of that HTML page which appears at the top of the browser window. It must be placed inside the head tag and should close immediately. (Optional)

<body> : Text between body tag describes the body content of the page that is visible to the end user. This tag contains the main content of the HTML document.

<h1> : Text between <h1> tag describes the first level heading of the webpage.

<p> : Text between <p> tag describes the paragraph of the webpage.

Features of HTML

1) It is a very easy and simple language. It can be easily understood and modified.
2) It is very easy to make an **effective presentation** with HTML because it has a lot of formatting tags.
3) It is a **markup language**, so it provides a flexible way to design web pages along with the text.
4) It facilitates programmers to add a **link** on the web pages (by html anchor tag), so it enhances the interest of browsing of the user.
5) It is **platform-independent** because it can be displayed on any platform like Windows, Linux, and Macintosh, etc.
6) It facilitates the programmer to add **Graphics, Videos, and Sound** to the web pages which makes it more attractive and interactive.
7) HTML is a case-insensitive language, which means we can use tags either in lower-case or upper-case.

HTML Tags

HTML tags are like keywords which defines that how web browser will format and display the content. With the help of tags, a web browser can distinguish between an HTML content and a simple content. HTML tags contain three main parts: opening tag, content and closing tag. But some HTML tags are unclosed tags.

When a web browser reads an HTML document, browser reads it from top to bottom and left to right. HTML tags are used to create HTML documents and render their properties. Each HTML tags have different properties.

An HTML file must have some essential tags so that web browser can differentiate between a simple text and HTML text. You can use as many tags you want as per your code requirement.

All HTML tags must enclosed within < > these brackets.

Every tag in HTML perform different tasks.

If you have used an open tag <tag>, then you must use a close tag </tag> (except some tags)

Syntax

<tag> content </tag>
 Paragraph Tag
<h2> Heading Tag </h2>
 Bold Tag
 Italic Tag
<u> Underline Tag</u>

Unclosed HTML Tags

Some HTML tags are not closed, for example br and hr.

 Tag: br stands for break line, it breaks the line of the code.
<hr> Tag: hr stands for Horizontal Rule. This tag is used to put a line across the webpage.

HTML Meta Tags

DOCTYPE, title, link, meta and style

HTML Text Tags

, <h1>, <h2>, <h3>, <h4>, <h5>, <h6>, , , <abbr>, <acronym>, <address>, <bdo>, <blockquote>, <cite>, <q>, <code>, <ins>, <del>, <dfn>, <kbd>, <pre>, <samp>, <var> and

HTML Link Tags

<a> and <base>

HTML Image and Object Tags

<img>, <area>, <map>, <param> and <object>
HTML List Tags
<ul>, <ol>, <li>, <dl>, <dt> and <dd>

HTML Table Tags

table, tr, td, th, tbody, thead, tfoot, col, colgroup and caption

HTML Form Tags

form, input, textarea, select, option, optgroup, button, label, fieldset and legend
HTML Scripting Tags
script and noscript

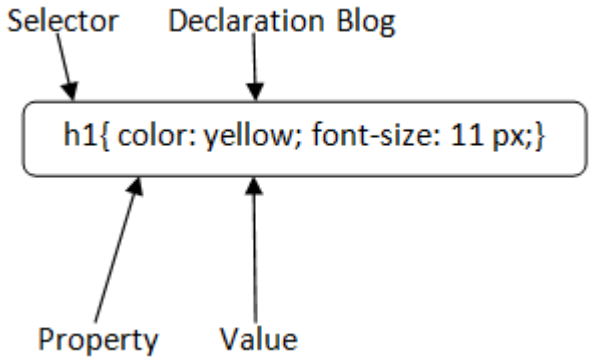
What is CSS?
CSS stands for Cascading Style Sheets. It is a style sheet language which is used to describe the look and formatting of a document written in markup language. It provides an additional feature to HTML. It is generally used with HTML to change the style of web pages and user interfaces. It can also be used with any kind of XML documents including plain XML, SVG and XUL.
CSS is used along with HTML and JavaScript in most websites to create user interfaces for web applications and user interfaces for many mobile applications.

What does CSS do

You can add new looks to your old HTML documents.
You can completely change the look of your website with only a few changes in CSS code.

Why use CSS

These are the three major benefits of CSS:
1) Solves a big problem
Before CSS, tags like font, color, background style, element alignments, border and size had to be repeated on every web page. This was a very long process. For example: If you are developing a large website where fonts and color information are added on every single page, it will be become a long and expensive process. CSS was created to solve this problem. It was a W3C recommendation.
2) Saves a lot of time
CSS style definitions are saved in external CSS files so it is possible to change the entire website by changing just one file.
3) Provide more attributes
CSS provides more detailed attributes than plain HTML to define the look and feel of the website.
CSS Syntax
A CSS rule set contains a selector and a declaration block.



Selector: Selector indicates the HTML element you want to style. It could be any tag like <h1>, <title> etc.
Declaration Block: The declaration block can contain one or more declarations separated by a semicolon. For the above example, there are two declarations:
color: yellow;
font-size: 11 px;
Each declaration contains a property name and value, separated by a colon.
Property: A Property is a type of attribute of HTML element. It could be color, border etc.
Value: Values are assigned to CSS properties. In the above example, value "yellow" is assigned to color property.
Selector{Property1: value1; Property2: value2; .....;}
CSS Selector
CSS selectors are used to select the content you want to style. Selectors are the part of CSS rule set. CSS selectors select HTML elements according to its id, class, type, attribute etc.
There are several different types of selectors in CSS.

1) CSS Element Selector
2) CSS Id Selector
3) CSS Class Selector
4) CSS Universal Selector
5) CSS Group Selector

1) CSS Element Selector
The element selector selects the HTML element by name.

```
<!DOCTYPE html>
<html>
<head>
<style>
p{
   text-align: center;
   color: blue;
}
</style>
</head>
<body>
<p>This style will be applied on every paragraph.</p>
<p id="para1">Me too!</p>
<p>And me!</p>
</body>
</html>
```

2) CSS Id Selector

The id selector selects the id attribute of an HTML element to select a specific element. An id is always unique within the page so it is chosen to select a single, unique element.

It is written with the hash character (#), followed by the id of the element.

Let?s take an example with the id "para1".

```
<!DOCTYPE html>
<html>
<head>
<style>
#para1 {
   text-align: center;
   color: blue;
}
</style>
</head>
<body>
<p id="para1">Hello Javatpoint.com</p>
<p>This paragraph will not be affected.</p>
</body>
</html>
```

3) CSS Class Selector

The class selector selects HTML elements with a specific class attribute. It is used with a period character . (full stop symbol) followed by the class name.

```
<!DOCTYPE html>
<html>
<head>
<style>
.center {
   text-align: center;
   color: blue;
}
</style>
</head>
<body>
<h1 class="center">This heading is blue and center-aligned.</h1>
<p class="center">This paragraph is blue and center-aligned.</p>
</body>
</html>
```

4) CSS Universal Selector

The universal selector is used as a wildcard character. It selects all the elements on the pages.

```
<!DOCTYPE html>
<html>
<head>
<style>
* {
  color: green;
  font-size: 20px;
```

```
}
</style>
</head>
<body>
<h2>This is heading</h2>
<p>This style will be applied on every paragraph.</p>
<p id="para1">Me too!</p>
<p>And me!</p>
</body>
</html>
```

5) CSS Group Selector

The grouping selector is used to select all the elements with the same style definitions.
Grouping selector is used to minimize the code. Commas are used to separate each selector in grouping.
Let's see the CSS code without group selector.

```
h1 {
   text-align: center;
   color: blue;
}
h2 {
   text-align: center;
   color: blue;
}
p {
   text-align: center;
   color: blue;
}
```

As you can see, you need to define CSS properties for all the elements. It can be grouped in following ways:

```
h1,h2,p {
   text-align: center;
   color: blue;
}
```

Let's see the full example of CSS group selector.

```
<!DOCTYPE html>
<html>
<head>
<style>
h1, h2, p {
   text-align: center;
   color: blue;
}
</style>
</head>
<body>
<h1>Hello Javatpoint.com</h1>
<h2>Hello Javatpoint.com (In smaller font)</h2>
<p>This is a paragraph.</p>
</body>
</html>
```

Conclusion: Thus Students have studied different HTML tags and their use. Students have also studied the different CSS selectors used along with HTML.

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Assignment No.3
HTML, CSS and JavaScript

Title: Implement an application in Java Script using following:
a) Design UI of application using HTML, CSS etc.
b) Include Java script validation
c) Use of prompt and alert window using Java Script
e.g., Design and implement a simple calculator using Java Script for operations like addition, multiplication, subtraction, division, square of number etc.
a) Design calculator interface like text field for input and output, buttons for numbers and operators etc.
b) Validate input values
c) Prompt/alerts for invalid values etc.

Theory:

What is JavaScript?

JavaScript (js) is a light-weight object-oriented programming language which is used by several websites for scripting the webpages. It is an interpreted, full-fledged programming language that enables dynamic interactivity on websites when applied to an HTML document. It was introduced in the year 1995 for adding programs to the webpages in the Netscape Navigator browser. Since then, it has been adopted by all other graphical web browsers. With JavaScript, users can build modern web applications to interact directly without reloading the page every time. The traditional website uses js to provide several forms of interactivity and simplicity.
Although, JavaScript has no connectivity with Java programming language. The name was suggested and provided in the times when Java was gaining popularity in the market. In addition to web browsers, databases such as CouchDB and MongoDB uses JavaScript as their scripting and query language.

Features of JavaScript

There are following features of JavaScript:

- All popular web browsers support JavaScript as they provide built-in execution environments.
- JavaScript follows the syntax and structure of the C programming language. Thus, it is a structured programming language.
- JavaScript is a weakly typed language, where certain types are implicitly cast (depending on the operation).
- JavaScript is an object-oriented programming language that uses prototypes rather than using classes for inheritance.
- It is a light-weighted and interpreted language.
- It is a case-sensitive language.
- JavaScript is supportable in several operating systems including, Windows, macOS, etc.
- It provides good control to the users over the web browsers.

History of JavaScript

In 1993, **Mosaic**, the first popular web browser, came into existence. In the **year 1994**, **Netscape** was founded by **Marc Andreessen**. He realized that the web needed to become more dynamic. Thus, a 'glue language' was believed to be provided to HTML to make web designing easy for designers and part-time programmers. Consequently, in 1995, the company recruited **Brendan Eich** intending to implement and embed Scheme programming language to the browser. But, before Brendan could start, the company merged with **Sun Microsystems** for adding Java into its Navigator so that it could compete with Microsoft over the web technologies and platforms. Now, two languages were there: Java and the scripting language. Further, Netscape decided to give a similar name to the scripting language as Java's. It led to 'Javascript'. Finally, in May 1995, Marc Andreessen coined the first code of Javascript named '**Mocha**'.

Later, the marketing team replaced the name with '**LiveScript**'. But, due to trademark reasons and certain other reasons, in December 1995, the language was finally renamed to 'JavaScript'. From then, JavaScript came into existence.

Application of JavaScript

JavaScript is used to create interactive websites. It is mainly used for:

- Client-side validation,
- Dynamic drop-down menus,
- Displaying date and time,
- Displaying pop-up windows and dialog boxes (like an alert dialog box, confirm dialog box and prompt dialog box),
- Displaying clocks etc.

JavaScript Example

```
<script>
document.write("Hello JavaScript by JavaScript");
</script>
```

HTML JavaScript

A Script is a small program which is used with HTML to make web pages more attractive, dynamic and interactive, such as an alert popup window on mouse click. Currently, the most popular scripting language is JavaScript used for websites.

Example:

```
<!DOCTYPE html>
<html>
<body>
<h1>JavaScript Date and Time example</h1>
<button type="button"
onclick="document.getElementById('demo').innerHTML = Date()">
Click me to display Date and Time.</button>
<p id="demo"></p>
</body>
</html>
```

HTML <script> Tag

The HTML <script> tag is used to specify a client-side script. It may be an internal or external JavaScript which contains scripting statements, hence we can place <script> tag within <body> or <head> section. It is mainly used to manipulate images, form validation and change content dynamically. JavaScript uses document.getElementById() method to select an HTML element.

Example:

```
<!DOCTYPE html>
<html>
<body>
<h2>Use JavaScript to Change Text</h2>
<p id="demo"></p>
<script>
document.getElementById("demo").innerHTML = "Hello JavaTpoint";
</script>
</body>
</html>
```

HTML events with JavaScript

An event is something which user does, or browser does such as mouse click or page loading are examples of events, and JavaScript comes in the role if we want something to happen on these events.

HTML provides event handler attributes which work with JavaScript code and can perform some action on an event.

Syntax:

```
<element event = "JS code">
```

Example:

```
<input type="button" value="Click" onclick="alert('Hi, how are you')">
```

Form events: reset, submit, etc.
Select events: text field, text area, etc.
Focus event: focus, blur, etc.
Mouse events: select, mouseup, mousemove, mousedown, click, dblclick, etc.
Following are the list for Window event attributes:

Event Event Name	Handler Name	Occurs when
onBlur	blur	When form input loses focus
onClick	click	When the user clicks on a form element or a link
onSubmit	submit	When user submits a form to the server.
onLoad	load	When page loads in a browser.
onFocus	focus	When user focuses on an input field.
onSelect	select	When user selects the form input filed.

1) JavaScript can change HTML content.

Example:

```
<!DOCTYPE html>
<html>
<body>
<p>JavaScript can change the content of an HTML element:</p>
<button type="button" onclick="myFunction()">Click Me!</button>
<p id="demo"></p>
<script>
function myFunction() {
   document.getElementById("demo").innerHTML = "Hello JavaTpoint!";
}
</script>
</body>
</html>
```

2) JavaScript can change HTML style

Example:

```
<!DOCTYPE html>
```

```
<html>
<body>
<p id="demo">JavaScript can change the style of an HTML element.</p>
<script>
function myFunction() {
   document.getElementById("demo").style.fontSize = "25px";
   document.getElementById("demo").style.color = "brown";
   document.getElementById("demo").style.backgroundColor = "lightgreen";
}
</script>
<button type="button" onclick="myFunction()">Click Me!</button>
</body>
</html>
```

3) JavaScript can change HTML attributes.

Example:

```
<!DOCTYPE html>
<html>
<body>
<script>
function light(sw) {
   var pic;
   if (sw == 0) {
      pic = "pic_lightoff.png"
   } else {
      pic = "pic_lighton.png"
   }
   document.getElementById('myImage').src = pic;
}
</script>
<img id="myImage" src="pic_lightoff.png" width="100" height="180">
<p>
<button type="button" onclick="light(1)">Light On</button>
<button type="button" onclick="light(0)">Light Off</button>
</p>
</body>
</html>
```

Use External Script

Suppose, you have various HTML files which should have same script, then we can put our JavaScript code in separate file and can call in HTML file. Save JavaScript external files using .js extension.

```
<!DOCTYPE html>
<html>
  <head>
   <script type="text/javascript" src="external.js"></script>
   </head>
    <body>
     <h2>External JavaScript Example</h2>
      <form onsubmit="fun()">
```

```
      <label>Enter your name:</label><br>
    <input type="text" name="uname" id="frm1"><br>
     <label>Enter your Email-address:</label><br>
    <input type="email" name="email"><br>
    <input type="submit">
 </form>
</body>
</html>

function fun() {
      var x = document.getElementById("frm1").value;
       alert("Hi"+" "+x+ "you have successfully submitted the details");
   }
```

Conclusion: Thus Students have studied what is JavaScript? And How it is used for the validations along with the HTML and CSS.

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Assignment No.4
Angular JS

Title: Design an application using Angular JS.
e.g., Design registration (first name, last name, username, password) and login page using Angular JS.

Theory:

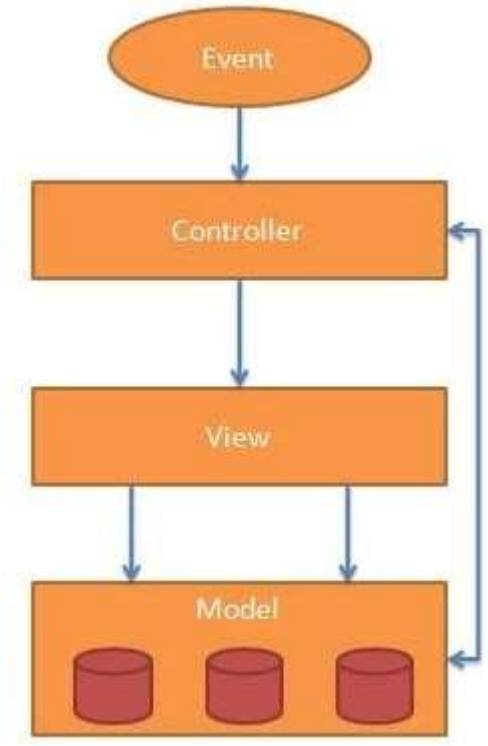
What is AngularJS?
Angular JS is an open source JavaScript framework that is used to build web applications. It can be freely used, changed and shared by anyone.
Angular Js is developed by Google.
It is an excellent framework for building single phase applications and line of business applications.

Advantage of AngularJS

There are a lot of JavaScript frameworks for building web applications. So, it is a genuine question, why to use Angular JS.

- Dependency Injection: Dependency Injection specifies a design pattern in which components are given their dependencies instead of hard coding them within the component.
- Two way data binding: AngularJS creates a two way data-binding between the select element and the orderProp model. orderProp is then used as the input for the orderBy filter.
- Testing: Angular JS is designed in a way that we can test right from the start. So, it is very easy to test any of its components through unit testing and end-to-end testing.
- Model View Controller: In Angular JS, it is very easy to develop application in a clean MVC way. You just have to split your application code into MVC components i.e. Model, View and the Controller.
- Directives, filters, modules, routes etc.

AngularJS MVC Architecture
MVC stands for Model View Controller. It is a software design pattern for developing web applications. It is very popular because it isolates the application logic from the user interface layer and supports separation of concerns.



The MVC pattern is made up of the following three parts:

- Model: It is responsible for managing application data. It responds to the requests from view and to the instructions from controller to update itself.
- View: It is responsible for displaying all data or only a portion of data to the users. It also specifies the data in a particular format triggered by the controller's decision to present the data. They are script-based template systems such as JSP, ASP, PHP and very easy to integrate with AJAX technology.
- Controller: It is responsible to control the relation between models and views. It responds to user input and performs interactions on the data model objects. The controller receives input, validates it, and then performs business operations that modify the state of the data model.

AngularJS First Example

AngularJS applications are a mix of HTML and JavaScript. The first thing you need is an HTML page.

```
<!DOCTYPE html>
<html>
<head>
.
.
</head>
<body>
.
.
</body>
</html>
```

Second, you need to include the AngularJS JavaScript file in the HTML page so we can use AngularJS:

```
<!DOCTYPE html>
<html>
```

```
<head>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.2.5/angular.min.js"></script>
</head>
<body>
.
.
</body>
</html>
```

AngularJS First Example

Following is a simple "Hello Word" example made with AngularJS. It specifies the Model, View, Controller part of an AngularJS app.

```
<!DOCTYPE html>
<html lang="en">
<head>
   <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.2.5/angular.min.js"></script>
</head>
<body ng-app="myapp">
<div ng-controller="HelloController" >
<h2>Hello {{helloTo.title}} !</h2>
</div>

<script>
angular.module("myapp", [])
   .controller("HelloController", function($scope) {
      $scope.helloTo = {};
      $scope.helloTo.title = "World, AngularJS";
   } );
</script>
</body>
</html>
```

View Part

```
<div ng-controller="HelloController" >
<h2>Hello {{helloTo.title}} !</h2>
</div>
```

Controller Part

```
<script>
angular.module("myapp", [])
   .controller("HelloController", function($scope) {
      $scope.helloTo = {};
      $scope.helloTo.title = "World, AngularJS";
   });
</script>
```

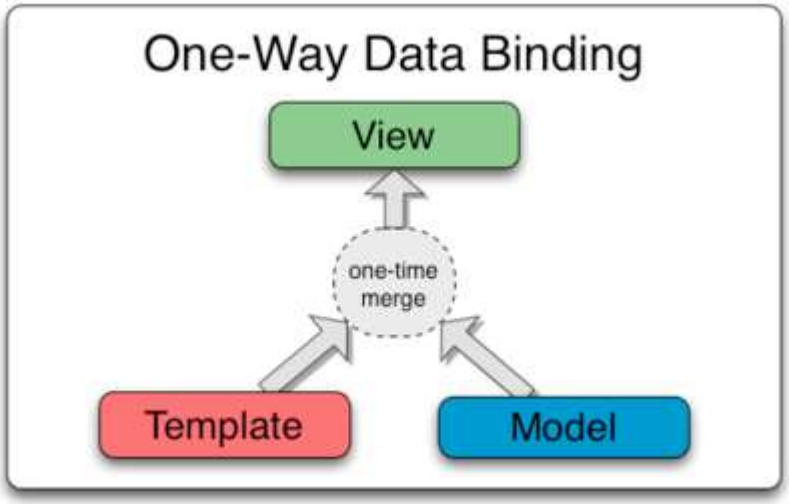
AngularJS Data Binding

Data binding is a very useful and powerful feature used in software development technologies. It acts as a bridge between the view and business logic of the application.

AngularJS follows Two-Way data binding model.

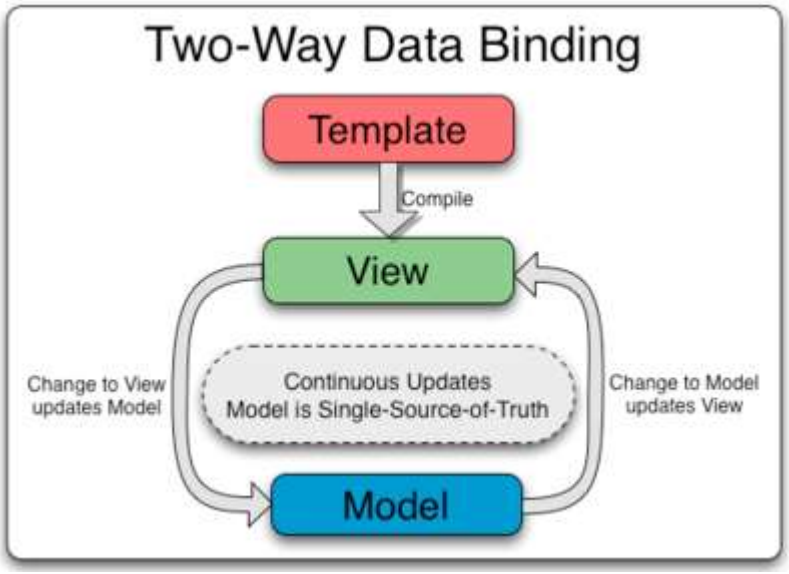
One-Way Data Binding

The one-way data binding is an approach where a value is taken from the data model and inserted into an HTML element. There is no way to update model from view. It is used in classical template systems. These systems bind data in only one direction.



Two-Way Data Binding

Data-binding in Angular apps is the automatic synchronization of data between the model and view components.
Data binding lets you treat the model as the single-source-of-truth in your application. The view is a projection of the model at all times. If the model is changed, the view reflects the change and vice versa.




```
<!DOCTYPE html>
<html>
<script src="http://ajax.googleapis.com/ajax/libs/angularjs/1.4.8/angular.min.js"></script>
<body>
<div ng-app="" ng-init="firstName='Ajeet'">
<p>Input something in the input box:</p>
```

```
<p>Name: <input type="text" ng-model="firstName"></p>
<p>You wrote: {{ firstName }}</p>
</div>
</body>
</html>
```

In the above example, the {{ firstName }} expression is an AngularJS data binding expression. Data binding in AngularJS binds AngularJS expressions with AngularJS data.
{{ firstName }} is bound with ng-model="firstName".
Let's take another example where two text fields are bound together with two ng-model directives:

```
<!DOCTYPE html>
<html>
<script src="http://ajax.googleapis.com/ajax/libs/angularjs/1.4.8/angular.min.js"></script>
<body>
<div data-ng-app="" data-ng-init="quantity=1;price=20">
<h2>Cost Calculator</h2>
Quantity: <input type="number" ng-model="quantity">
Price: <input type="number" ng-model="price">
<p><b>Total in rupees:</b> {{quantity * price}}</p>
</div>
</body>
</html>
```

AngularJS Expressions

In AngularJS, expressions are used to bind application data to HTML. AngularJS resolves the expression, and return the result exactly where the expression is written.
Expressions are written inside double braces {{expression}}.They can also be written inside a directive: ng-bind="expression".
AnularJS expressions are very similar to JavaScript expressions. They can contain literals, operators, and variables. For example:
{{ 5 + 5 }} or {{ firstName + " " + lastName }}

AngularJS Expressions Example

```
<!DOCTYPE html>
<html>
<script src="http://ajax.googleapis.com/ajax/libs/angularjs/1.4.8/angular.min.js"></script>
<body>
<div ng-app>
<p>A simple expression example: {{ 5 + 5 }}</p>
</div>
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<script src="http://ajax.googleapis.com/ajax/libs/angularjs/1.4.8/angular.min.js"></script>
<body>
<p>If you remove the directive "ng-app", HTML will display the expression without solving it.</p>
<div>
<p>A simple expression example:  {{ 5 + 5 }}</p>
```

```
</div>
</body>
</html>

<!DOCTYPE html>
<html>
<script src="http://ajax.googleapis.com/ajax/libs/angularjs/1.4.8/angular.min.js"></script>
<body>
<p>Change the value of the input field:</p>
<div ng-app="" ng-init="myCol='pink'">
<input style="background-color:{{myCol}}" ng-model="myCol" value="{{myCol}}">
</div>
<p>AngularJS resolves the expression and returns the result.</p>
<p>The background color of the input box will be whatever you write in the input field.</p>
</body>
</html>
```

Conclusion: Thus Students have studied what is Angular JS? And how it is used for the development of web application.

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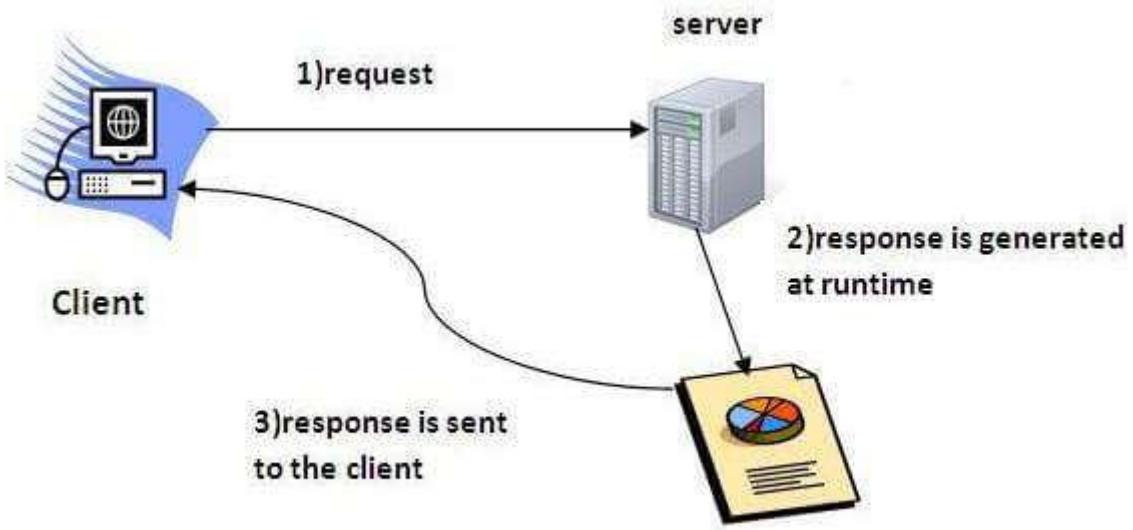
Assignment No.5
Servlet

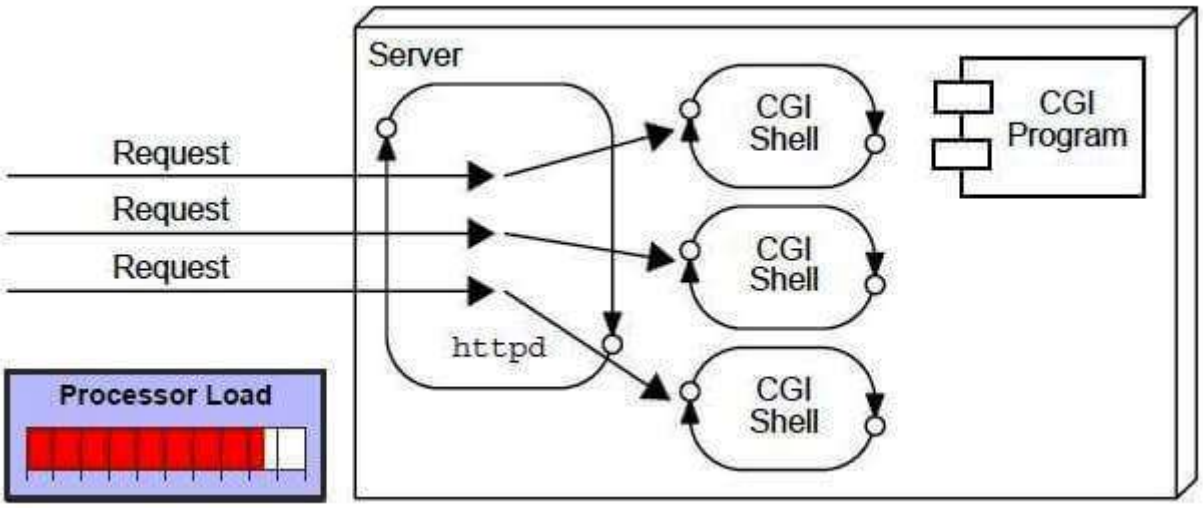
Title: Implement the sample program demonstrating the use of Servlet.
e.g., Create a database table ebookshop (book_id, book_title, book_author, book_price, quantity) using database like Oracle/MySQL etc. update and display (use SQL select query) the table content using servlet.

Theory:

What is a Servlet?
Servlet can be described in many ways, depending on the context.
Servlet is a technology which is used to create a web application.
Servlet is an API that provides many interfaces and classes including documentation.
Servlet is an interface that must be implemented for creating any Servlet.
Servlet is a class that extends the capabilities of the servers and responds to the incoming requests. It can respond to any requests.
Servlet is a web component that is deployed on the server to create a dynamic web page.



What is a web application?
A web application is an application accessible from the web. A web application is composed of web components like Servlet, JSP, Filter, etc. and other elements such as HTML, CSS, and JavaScript. The web components typically execute in Web Server and respond to the HTTP request.
CGI (Common Gateway Interface)
CGI technology enables the web server to call an external program and pass HTTP request information to the external program to process the request. For each request, it starts a new process.



Disadvantages of CGI

There are many problems in CGI technology:

If the number of clients increases, it takes more time for sending the response.

For each request, it starts a process, and the web server is limited to start processes.

It uses platform dependent language e.g. C, C++, perl.

Advantages of Servlet

There are many advantages of Servlet over CGI. The web container creates threads for handling the multiple requests to the Servlet. Threads have many benefits over the Processes such as they share a common memory area, lightweight, cost of communication between the threads are low. The advantages of Servlet are as follows:

Better performance: because it creates a thread for each request, not process.

Portability: because it uses Java language.

Robust: JVM manages Servlets, so we don't need to worry about the memory leak, garbage collection, etc.

Secure: because it uses java language.

Servlet Interface

Servlet interface provides commonbehaviorto all the servlets.Servlet interface defines methods that all servlets must implement.

Servlet interface needs to be implemented for creating any servlet (either directly or indirectly). It provides 3 life cycle methods that are used to initialize the servlet, to service the requests, and to destroy the servlet and 2 non-life cycle methods.

Methods of Servlet interface

There are 5 methods in Servlet interface. The init, service and destroy are the life cycle methods of servlet. These are invoked by the web container.

GenericServlet class

GenericServlet class implements Servlet, ServletConfig and Serializable interfaces. It provides the implementation of all the methods of these interfaces except the service method.

GenericServlet class can handle any type of request so it is protocol-independent.

You may create a generic servlet by inheriting the GenericServlet class and providing the implementation of the service method.

Methods of GenericServlet class

public void init(ServletConfig config) is used to initialize the servlet.

public abstract void service(ServletRequest request, ServletResponse response) provides service for the incoming request. It is invoked at each time when user requests for a servlet.

public void destroy() is invoked only once throughout the life cycle and indicates that servlet is being destroyed.

public ServletConfig getServletConfig() returns the object of ServletConfig.
public String getServletInfo() returns information about servlet such as writer, copyright, version etc.
public void init() it is a convenient method for the servlet programmers, now there is no need to call super.init(config)
public ServletContext getServletContext() returns the object of ServletContext.
public String getInitParameter(String name) returns the parameter value for the given parameter name.
public Enumeration getInitParameterNames() returns all the parameters defined in the web.xml file.
public String getServletName() returns the name of the servlet object.
public void log(String msg) writes the given message in the servlet log file.
public void log(String msg,Throwable t) writes the explanatory message in the servlet log file and a stack trace.

```
import java.io.*;
import javax.servlet.*;

public class First extends GenericServlet{
public void service(ServletRequest req,ServletResponse res)
throws IOException,ServletException{

res.setContentType("text/html");

PrintWriter out=res.getWriter();
out.print("<html><body>");
out.print("<b>hello generic servlet</b>");
out.print("</body></html>");

}
}
```

HttpServlet class

The HttpServlet class extends the GenericServlet class and implements Serializable interface. It provides http specific methods such as doGet, doPost, doHead, doTrace etc.

Methods of HttpServlet class

There are many methods in HttpServlet class. They are as follows:

public void service(ServletRequest req,ServletResponse res) dispatches the request to the protected service method by converting the request and response object into http type.

protected void service(HttpServletRequest req, HttpServletResponse res) receives the request from the service method, and dispatches the request to the doXXX() method depending on the incoming http request type.

protected void doGet(HttpServletRequest req, HttpServletResponse res) handles the GET request. It is invoked by the web container.

protected void doPost(HttpServletRequest req, HttpServletResponse res) handles the POST request. It is invoked by the web container.

protected void doHead(HttpServletRequest req, HttpServletResponse res) handles the HEAD request. It is invoked by the web container.

protected void doOptions(HttpServletRequest req, HttpServletResponse res) handles the OPTIONS request. It is invoked by the web container.

protected void doPut(HttpServletRequest req, HttpServletResponse res) handles the PUT request. It is invoked by the web container.

protected void doTrace(HttpServletRequest req, HttpServletResponse res) handles the TRACE request. It is invoked by the web container.

protected void doDelete(HttpServletRequest req, HttpServletResponse res) handles the DELETE request. It

is invoked by the web container.
protected long getLastModified(HttpServletRequest req) returns the time when HttpServletRequest was last modified since midnight January 1, 1970 GMT.

Servlet Program:

```
import jakarta.servlet.http.*;
import jakarta.servlet.*;
import java.io.*;
import java.sql.*;
public class DemoServlet extends HttpServlet{
public void doGet(HttpServletRequest req,HttpServletResponse res)
throws ServletException,IOException
{
res.setContentType("text/html");//setting the content type
PrintWriter pw=res.getWriter();//get the stream to write the data
pw.println("<html><body>");
pw.println("Welcome to Pragati eBookShop");
pw.println("<table border='5'>");
pw.println("<tr><th>Book id</th><th>Book Title</th><th>Book Author</th><th>Book Price</th><th>Quantity</th></tr>");
try{
Class.forName("com.mysql.jdbc.Driver");

Connection con=DriverManager.getConnection("jdbc:mysql://localhost:3306/pragati","root","");

Statement stmt=con.createStatement();

ResultSet rs=stmt.executeQuery("select * from ebookshop");
while(rs.next()){
//writing html in the stream
pw.println("<tr><td>"+rs.getObject(1).toString()+"</td><td>"+rs.getString(2)+"</td><td>"+rs.getString(3)+"</td><td>"+rs.getDouble(4)+"</td><td>"+rs.getInt(5)+"</td></tr>");

}
}catch(Exception e){ pw.print(e);}
pw.println("</table>");
pw.println("</body></html>");
pw.close();//closing the stream
}}
```

Conclusion: Thus Students have studied what is Servlet? And how it is used for the development of web application.

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Assignment No.6
JSP

Title: Implement the program demonstrating the use of JSP.
e.g., Create a database table students_info (stud_id, stud_name, class, division, city) using database like Oracle/MySQL etc. update and display (use SQL select query) the table content using JSP.

Theory:

What is a JSP?

JSP technology is used to create web application just like Servlet technology. It can be thought of as an extension to Servlet because it provides more functionality than servlet such as expression language, JSTL, etc.

A JSP page consists of HTML tags and JSP tags. The JSP pages are easier to maintain than Servlet because we can separate designing and development. It provides some additional features such as Expression Language, Custom Tags, etc.

Advantages of JSP over Servlet

There are many advantages of JSP over the Servlet. They are as follows:

1) Extension to Servlet

JSP technology is the extension to Servlet technology. We can use all the features of the Servlet in JSP. In addition to, we can use implicit objects, predefined tags, expression language and Custom tags in JSP, that makes JSP development easy.

2) Easy to maintain

JSP can be easily managed because we can easily separate our business logic with presentation logic. In Servlet technology, we mix our business logic with the presentation logic.

3) Fast Development: No need to recompile and redeploy

If JSP page is modified, we don't need to recompile and redeploy the project. The Servlet code needs to be updated and recompiled if we have to change the look and feel of the application.

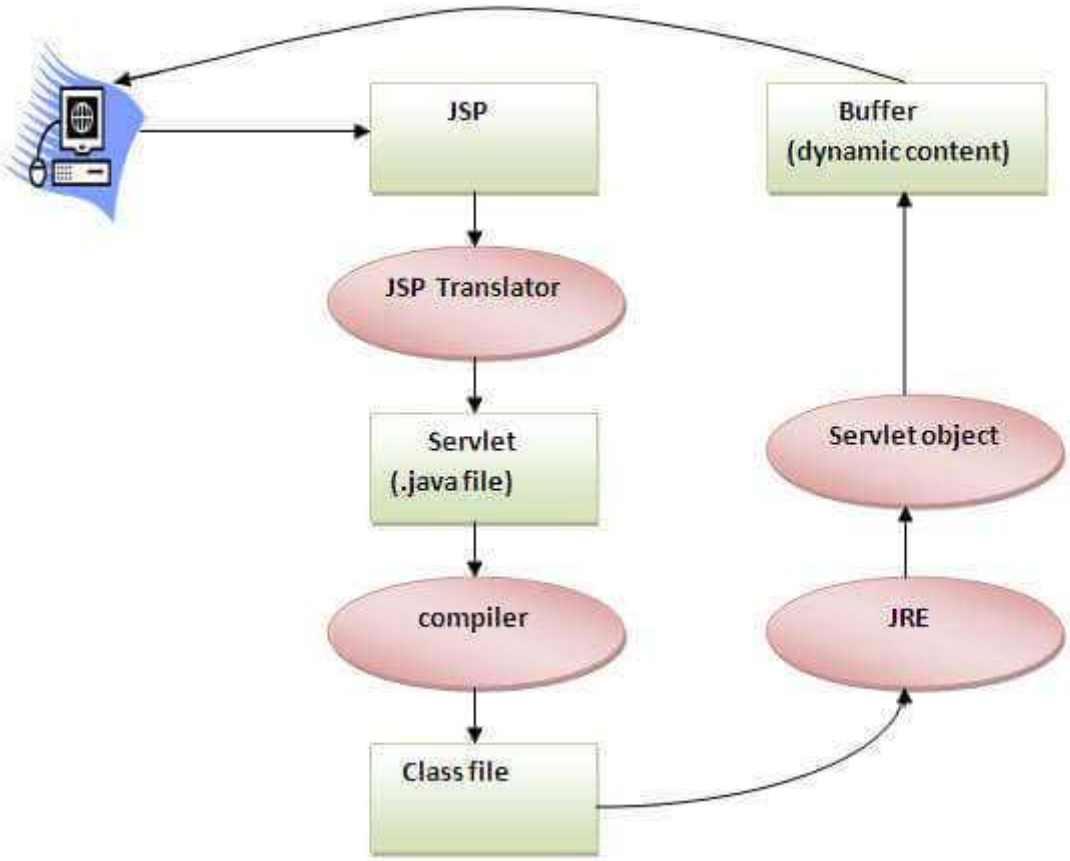
4) Less code than Servlet

In JSP, we can use many tags such as action tags, JSTL, custom tags, etc. that reduces the code. Moreover, we can use EL, implicit objects, etc.

The Lifecycle of a JSP Page

The JSP pages follow these phases:

- Translation of JSP Page
- Compilation of JSP Page
- Classloading (the classloader loads class file)
- Instantiation (Object of the Generated Servlet is created).
- Initialization (the container invokes jspInit() method).
- Request processing (the container invokes _jspService() method).
- Destroy (the container invokes jspDestroy() method).



As depicted in the above diagram, JSP page is translated into Servlet by the help of JSP translator. The JSP translator is a part of the web server which is responsible for translating the JSP page into Servlet. After that, Servlet page is compiled by the compiler and gets converted into the class file. Moreover, all the processes that happen in Servlet are performed on JSP later like initialization, committing response to the browser and destroy.

Creating a simple JSP Page

To create the first JSP page, write some HTML code as given below, and save it by .jsp extension. We have saved this file as index.jsp. Put it in a folder and paste the folder in the web-apps directory in apache tomcat to run the JSP page.

index.jsp

Let's see the simple example of JSP where we are using the scriptlet tag to put Java code in the JSP page. We will learn scriptlet tag later.

```
1. <html>
2. <body>
3. <% out.print(2*5); %>
4. </body>
5. </html>
```

It will print 10 on the browser.

How to run a simple JSP Page?

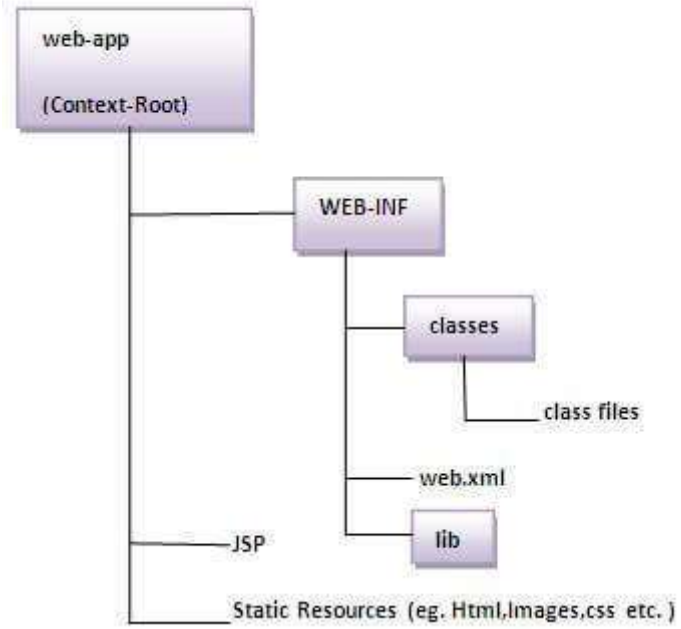
Follow the following steps to execute this JSP page:

- Start the server

- Put the JSP file in a folder and deploy on the server
- Visit the browser by the URL http://localhost:portno/contextRoot/jspfile, for example, http://localhost:8888/myapplication/index.jsp

The Directory structure of JSP

The directory structure of JSP page is same as Servlet. We contain the JSP page outside the WEB-INF folder or in any directory.




```
<%@ page import="java.sql.*"%>
<html>
<body>
<% out.print("Hello World !! Welcome to JSP"); %>
<p>Hello World !! Welcome to JSP</p>
<h1>Welcome to Sanjivani College of Engineering, IT Department</h1>
<table border='5'>
<tr><th>Stud_id</th><th>Name</th><th>Class</th><th>Division</th><th>City</th></tr>
<%
try{
Class.forName("com.mysql.jdbc.Driver");

Connection con=DriverManager.getConnection("jdbc:mysql://localhost:3306/pragati","root","");

Statement stmt=con.createStatement();

ResultSet rs=stmt.executeQuery("select * from student_info");
while(rs.next()){
//writing html in the stream
out.println("<tr><td>"+rs.getObject(1).toString()+"</td><td>"+rs.getString(2)+"</td><td>"+rs.getString(
3)+"</td><td>"+rs.getString(4)+"</td><td>"+rs.getString(5)+"</td></tr>");

}
}catch(Exception e){ out.print(e);}
```

```
%>
</table>
</body>
</html>
```

Conclusion: Thus Students have studied what is JSP? And how it is used for the development of web application.

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Assignment No.7
PHP and MySQL

Title: Build a dynamic web application using PHP and MySQL.
a. Create database tables in MySQL and create connection with PHP.
b. Create the add, update, delete and retrieve functions in the PHP web app interacting with MySQL database.

Theory:

What is PHP?

PHP is an open-source, interpreted, and object-oriented scripting language that can be executed at the server-side. PHP is well suited for web development. Therefore, it is used to develop web applications (an application that executes on the server and generates the dynamic page.).

PHP was created by Rasmus Lerdorf in 1994 but appeared in the market in 1995. PHP 7.4.0 is the latest version of PHP, which was released on 28 November. Some important points need to be noticed about PHP are as followed:

- PHP is an interpreted language, i.e., there is no need for compilation.
- PHP is faster than other scripting languages, for example, ASP and JSP.
- PHP is a server-side scripting language, which is used to manage the dynamic content of the website.
- PHP can be embedded into HTML.
- PHP is an object-oriented language.
- PHP is an open-source scripting language.
- PHP is simple and easy to learn language.

PHP Example

```
<?php
   echo "Hello World!";
?>
```

Why use PHP

PHP is a server-side scripting language, which is used to design the dynamic web applications with MySQL database.

It handles dynamic content, database as well as session tracking for the website.

You can create sessions in PHP.

It can access cookies variable and also set cookies.

It helps to encrypt the data and apply validation.

PHP supports several protocols such as HTTP, POP3, SNMP, LDAP, IMAP, and many more.

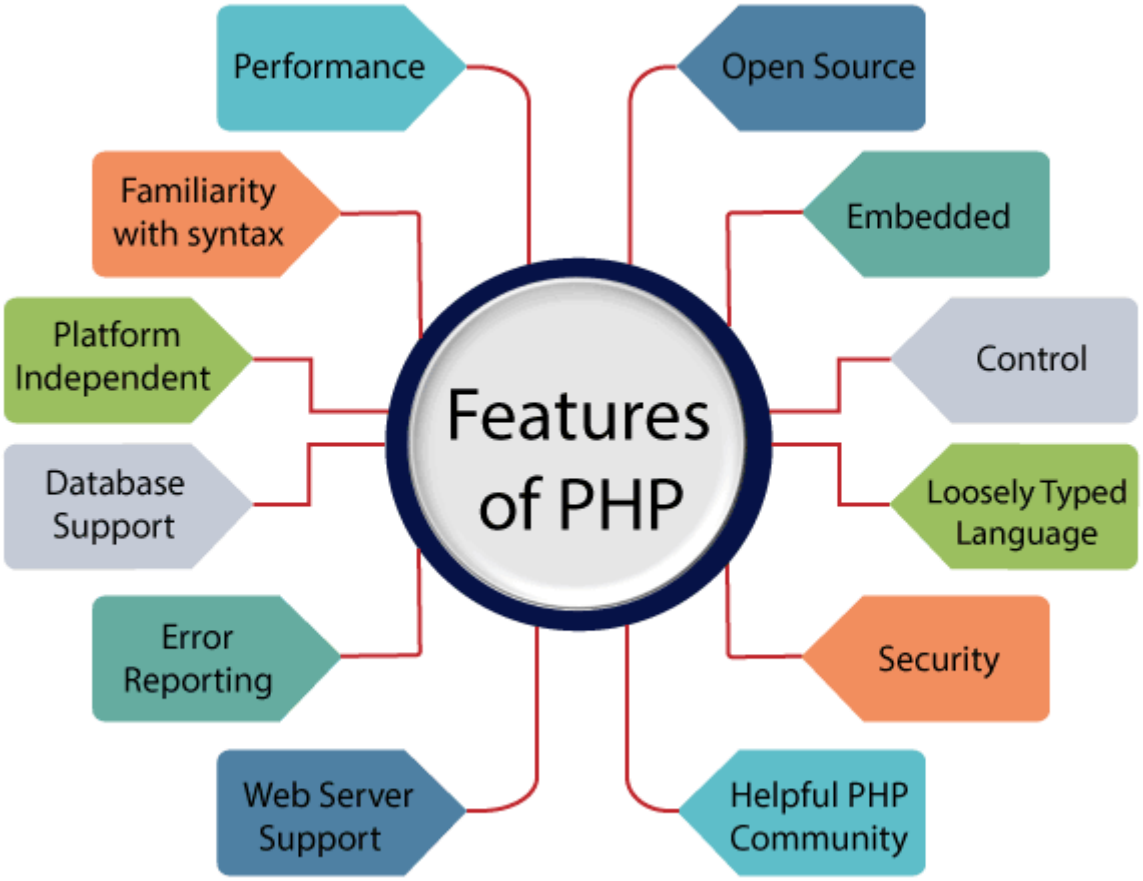
Using PHP language, you can control the user to access some pages of your website.

As PHP is easy to install and set up, this is the main reason why PHP is the best language to learn.

PHP can handle the forms, such as - collect the data from users using forms, save it into the database, and return useful information to the user. For example - Registration form.

PHP Features

PHP is very popular language because of its simplicity and open source. There are some important features of PHP given below:



Web Development

PHP is widely used in web development nowadays. PHP can develop dynamic websites easily. But you must have the basic the knowledge of following technologies for web development as well.

HTML
CSS
JavaScript
Ajax
XML and JSON
jQuery

addressbook.php

```
<html>
<body>
<h1>Address Book</h1>
<form name="f1" action="" method="post">
<table >
<tr><td>UserId:</td>
<td><input type="text" name="t1" value=""/></td></tr>
<tr><td>Name:</td>
<td><input type="text" name="t2" value=""/></td></tr>
<tr><td>PhoneNo:</td>
<td><input type="text" name="t3" value=""/></td></tr>
<tr><td>Email:</td>
<td><input type="text" name="t4" value=""/></td></tr>
```

```
<tr><td>Permanent Address:</td>
<td><input type="text" name="t5" value=""/></td></tr>
<tr><td>Temporary Address:</td>
<td><input type="text" name="t6" value=""/></td></tr>
</table>
<input type="submit" name="but1" value="ADD"/>
<input type="submit" name="but2" value="UPDATE"/>
<input type="submit" name="but3" value="DELETE"/>
<input type="submit" name="but4" value="DISPLAY"/>
</form>
<?php
if(isset($_POST['but1']))
{
$con=mysqli_connect("localhost","root","","addressbook");
// Check connection
if (mysqli_connect_errno()) {
  echo "Failed to connect to mysqli: " . mysqli_connect_error();
}
$userid = $_POST["t1"];
$name =  $_POST["t2"];
$phno = $_POST["t3"];
$email = $_POST["t4"];
$padd = $_POST["t5"];
$tadd = $_POST["t6"];
$sql="INSERT INTO person (id, name, phno, email, padd, tadd)
VALUES ('$userid', '$name', '$phno', '$email', '$padd', '$tadd')";

if (!mysqli_query($con,$sql)) {
  die('Error: ' . mysqli_error($con));
}
echo "1 record added successfully";

//mysqli_close($con);
}
else if(isset($_POST['but4']))
{
header('Location: display.php');
}
else if(isset($_POST['but3']))
{
header('Location: delete.html');
}
else if(isset($_POST['but2']))
{
header('Location: update.html');;
}
?>

</body>
</html>
```

Save.php

```
<?php
session_start();
$uid=$_SESSION['uid'];
$con=mysqli_connect("localhost","root","","addressbook");
// Check connection
if (mysqli_connect_errno()) {
  echo "Failed to connect to mysqli: " . mysqli_connect_error();
}
$userid = $_POST["t1"];
$name =  $_POST["t2"];
$phno = $_POST["t3"];
$email = $_POST["t4"];
$padd = $_POST["t5"];
$tadd = $_POST["t6"];

$sql1="select * from person where id=$uid";
$result = mysqli_query($con,$sql1);
    if(mysqli_num_rows($result) == 1)
    {
    $sql="UPDATE person SET
id='$userid',name='$name',phno='$phno',email='$email',padd='$padd',tadd='$tadd' WHERE id=$uid";
      if (!mysqli_query($con,$sql)) {
      die('Error: ' . mysqli_error($con));
                                }
      else{
          echo "Record Updated Successfully";
         }
    }
           else{
           echo "Record not present";
           }

mysqli_close($con);
?>
<html>
<body>
<form name="f1" action="addressbook.php" method="post">
<input type="submit" name="display" value="OK"/>
</form>
</body>
</html>
```

Display.php

```
<?php
$username = "root";
$password = "";
$hostname = "localhost";

//connection to the database
```

```
$dbhandle = mysql_connect($hostname, $username, $password)
 or die("Unable to connect to MySQL");
echo "Connected to MySQL<br>";

//select a database to work with
$selected = mysql_select_db("addressbook",$dbhandle)
  or die("Could not select examples");

//execute the SQL query and return records
$result = mysql_query("SELECT  id, name, phno, email, padd, tadd FROM person");

//fetch tha data from the database
echo "<h1><center>AddressBook Information</center></h1>";
echo "<table border='5' align='center'>";
echo "<tr><th>ID</th><th>Name</th><th>Phone No</th><th>Email</th><th>P ADD</th><th>T
ADD</th></tr>";
while ($row = mysql_fetch_array($result)) {
   echo "<tr><td>".$row{'id'}."</td>";
   echo "<td>".$row{'name'}."</td>";
   echo "<td>".$row{'phno'}."</td>";
   echo "<td>".$row{'email'}."</td>";
   echo "<td>".$row{'padd'}."</td>";
   echo "<td>".$row{'tadd'}."</td></tr>";
}
echo "</table>";
//close the connection
mysql_close($dbhandle);
?>
<html>
<body>
<form name="f1" action="addressbook.php" method="post">
<input type="submit" name="display" value="back"/>
</form>
</body>
</html>
```

Delete.php

```
<?php
$con=mysqli_connect("localhost","root","","addressbook");
// Check connection
if (mysqli_connect_errno()) {
  echo "Failed to connect to mysqli: " . mysqli_connect_error();
}
//execute the SQL query and return records

$userid=$_POST['uid'];
$sql1="select * from person where id=$userid";
$result = mysqli_query($con,$sql1);
    if(mysqli_num_rows($result) == 1) // User not found. So, redirect to login_form again.
    {
```

```
   $sql="DELETE FROM person WHERE id=$userid";
     if (!mysqli_query($con,$sql)) {
     die('Error: ' . mysqli_error($con));
                                }
     else{
          echo "Record Deleted Successfully";
         }
   }
          else{
          echo "Record not present";
          }
mysqli_close($con);
?>
<html>
<body>
<form name="f1" action="addressbook.php" method="post"/>
<input type="submit" name="back" value="Back"/>
</form>
</body>
</html>
```

Conclusion: Thus Students have studied what is JSP? And how it is used for the development of web application.

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Assignment No.8
Struts

Title: Design a login page with entries for name, mobile number email id and login button. Use struts and perform following validations
a. Validation for correct names
b. Validation for mobile numbers
c. Validation for email id
d. Validation if no entered any value
e. Re-display for wrongly entered values with message
f. Congratulations and welcome page upon successful entries.

Theory:

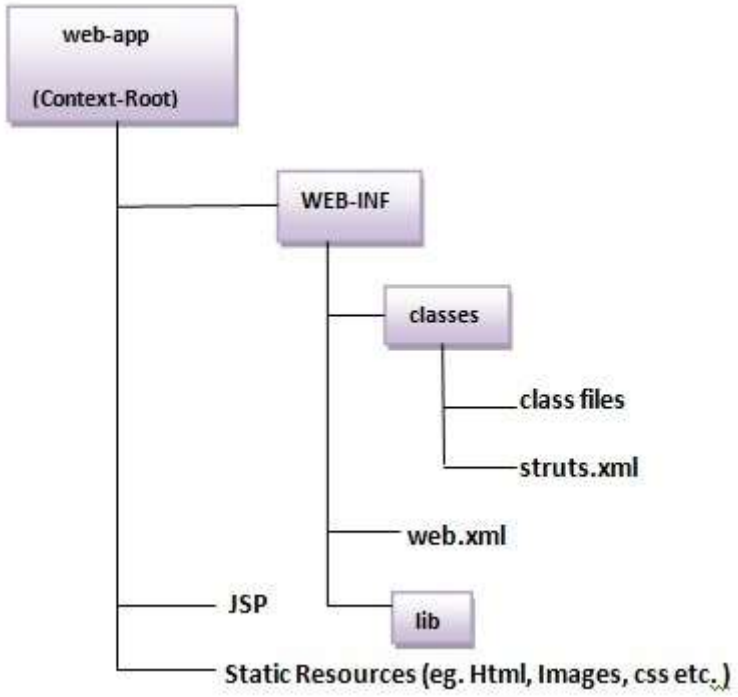
What is Struts?
Struts is a free, open-source Java framework that helps developers create web applications using the Model-View-Controller (MVC) architecture.

Steps to create Struts 2 Application Example

In this example, we are creating the struts 2 example without IDE. We can simply create the struts 2 application by following these simple steps:

1. Create the directory structure
2. Create input page (index.jsp)
3. Provide the entry of Controller in (web.xml) file
4. Create the action class (Product.java)
5. Map the request with the action in (struts.xml) file and define the view components
6. Create view components (welcome.jsp)
7. load the jar files
8. start server and deploy the project

1) Create the directory structure
The directory structure of struts 2 is same as servlet/JSP. Here, struts.xml file must be located in the classes folder.



2) Create input page (index.jsp)

This jsp page creates a form using struts UI tags. To use the struts UI tags, you need to specify uri /struts-tags. Here, we have used s:form to create a form, s:textfield to create a text field, s:submit to create a submit button.

index.jsp

```
1. <%@ taglib uri="/struts-tags" prefix="s" %>
2. <s:form action="product">
3. <s:textfield name="id" label="Product Id"></s:textfield>
4. <s:textfield name="name" label="Product Name"></s:textfield>
5. <s:textfield name="price" label="Product Price"></s:textfield>
6. <s:submit value="save"></s:submit>
7. </s:form>
```

3) Provide the entry of Controller in (web.xml) file

In struts 2, StrutsPrepareAndExecuteFilter class works as the controller. As we know well, struts 2 uses filter for the controller. It is implicitly provided by the struts framework.

web.xml

```
1. <?xml version="1.0" encoding="UTF-8"?>
2. <web-app>
3.   <filter>
4.   <filter-name>struts2</filter-name>
5.    <filter-class>
6.     org.apache.struts2.dispatcher.ng.filter.StrutsPrepareAndExecuteFilter
7.    </filter-class>
8.   </filter>
9.   <filter-mapping>
10.   <filter-name>struts2</filter-name>
11.    <url-pattern>/*</url-pattern>
12.  </filter-mapping>
```

```
13. </web-app>
```

4) Create the action class (Product.java)

This is simple bean class. In struts 2, action is POJO (Plain Old Java Object). It has one extra method execute i.e. invoked by struts framework by default.

Product.java

```
1. package com.javatpoint;
2.
3. public class Product {
4. private int id;
5. private String name;
6. private float price;
7. public int getId() {
8.     return id;
9. }
10. public void setId(int id) {
11.     this.id = id;
12. }
13. public String getName() {
14.     return name;
15. }
16. public void setName(String name) {
17.     this.name = name;
18. }
19. public float getPrice() {
20.     return price;
21. }
22. public void setPrice(float price) {
23.     this.price = price;
24. }
25.
26. public String execute(){
27.     return "success";
28. }
29. }
```

5) Map the request in (struts.xml) file and define the view components

It is the important file from where struts framework gets information about the action and decides which result to be invoked. Here, we have used many elements such as struts, package, action and result.

struts element is the root elements of this file. It represents an application.

package element is the sub element of struts. It represents a module of the application. It generally extends the struts-default package where many interceptors and result types are defined.

action element is the sub element of package. It represents an action to be invoked for the incoming request. It has name, class and method attributes. If you don't specify name attribute by default execute() method will be invoked for the specified action class.

result element is the sub element of action. It represents an view (result) that will be invoked. Struts framework checks the string returned by the action class, if it returns success, result page for the action is invoked whose name is success or has no name. It has name and type attributes. Both are optional. If you

don't specify the result name, by default success is assumed as the result name. If you don't specify the type attribute, by default dispatcher is considered as the default result type. We will learn about result types later.

struts.xml

```
<?xml version="1.0" encoding="UTF-8" ?>
<!DOCTYPE struts PUBLIC "-//Apache Software Foundation//DTD Struts
Configuration 2.1//EN" "http://struts.apache.org/dtds/struts-2.1.dtd">
<struts>
<package name="default" extends="struts-default">

<action name="product" class="com.javatpoint.Product">
<result name="success">welcome.jsp</result>
</action>

</package>
</struts>
```

6) Create view components (welcome.jsp)

It is the view component the displays information of the action. Here, we are using struts tags to get the information.

The s:property tag returns the value for the given name, stored in the action object.

welcome.jsp

```
<%@ taglib uri="/struts-tags" prefix="s" %>

Product Id:<s:property value="id"/><br/>
Product Name:<s:property value="name"/><br/>
Product Price:<s:property value="price"/><br/>
```

7) Load the jar files

To run this application, you need to have the struts 2 jar files. Here, we are providing all the necessary jar files for struts 2. Download it and put these jar files in the lib folder of your project.

8) start server and deploy the project

Finally, start the server and deploy the project and access it.

Conclusion: Thus Students have studied what is Struts? And how it is used for the development of web application.

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Assignment No.9
Wordpress

Title: Design a website using Content management system of WordPress. Make the use of different plugins and themes of the WordPress.

Theory:

What is WordPress?

WordPress is a free and open-source **C**ontent **M**anagement **S**ystem (CMS). It is an online site based on PHP and MySQL. It was mainly designed as a blogging tool but now it has evolved into a versatile CMS. Where, you can use it to create a simple blog as well as a fully operationable website and mobile applications. It is even used to create an online store using some WordPress plugins.

But some people still misunderstand it just as a blogging platform.

It is considered as the easiest and most popular CMS tool due to its features. The main feature of WordPress is its versatility and feasibility to use. There is no use of coding and designing skills for creating a website on this. Even a non-technical person can also create a website with the help of WordPress easily.

WordPress is an open-source community that implies thousands of people from all over the world work on it. It is free software. You are free to download, install, modify and use it. Although, there might be some cost involved during web hosting.

Reasons to use WordPress

There are many reasons to use WordPress in today's scenario as it provides a great help to its users in all respect.

Creating a website without logical skill

The most interesting and best part about WordPress is that anyone can create his/her own website on this without any coding or designing skills. Out of million users on WordPress, most of them are neither designers nor programmers.

To create a website on WordPress you only need an Internet connection and a web browser.

Provide a range of Themes

It gives thousand types of template options to give any type of feel and look to a website. They are very easy to customize as they come with their own option panel which allows users to select colors, background, sliders, fonts, logos and many more.

Plugins to add functionality

It provides both free and premium types of plugins. These plugins add extra functionality or may give a whole new platform to a website. It allows a user to add photo galleries, shopping carts and much more.

Free and Open-source platform

WordPress is a free and open-source software. It is free to install and use. As a new user you can easily create a website of your own that too absolutely free. It never has any type of hidden charges once a user started using it.

Being an open-source, you can alter the source-code according to your need. Anyone can become a contributor to WordPress by answering questions, creating themes or writing plugins.

Adding a blog is very easy

WordPress came into existence with blogging and still mostly consider it as a blogging site. It provides all type of blog solutions from designing, styling to the anti spam solutions. It will meet your demands in all possible ways.

More secure

It only takes 5 minutes to install, and is really fast in updating for security and new functionality. Upgrading to a new version is automatic and very fast as it provides a one link click upgradation. It constantly updates its site and software to prevent from hacking.

Google and WordPress

In a press conference in 2009, Matt Cutts, the head of Google's web spam team said that "Google loves WordPress". Even WordPress simple plugins and themes are also really attractive to Google. WordPress is a better option when doing SEO due to its framework which is easy to crawl.

Inserting Multimedia

A user whether writing a blog or creating a website, need to insert videos, pictures to make its content more powerful. WordPress makes it very easy to upload a multimedia file or to make some editing in files like images.

Mobile-Friendly

A website needs to be responsive for small screens like tablets, smartphones to reach all its users. WordPress is mobile-friendly as its themes are made responsive.

Easy to use

WordPress dashboard always look the same. You don't need to hire an expert to learn how to use WordPress.

Installing WordPress

Installing WordPress is very easy and takes less than five minutes to complete. It is best known for its ease of installation. All web hosting companies allow WordPress installation within few clicks.

There are different methods to install WordPress on a system.

Methods to install WordPress

Using cPanel

This is the fastest and easiest way to install WordPress on your system using cPanel software. The most common tools available to automatically install WordPress are Softaculous, SimpleScripts, Fantastico and De Luxe. All the web hosts allow one-click installation through these tools.

You only have to choose one WordPress host and use installation script in cPanel. You will get your login details and URL link of cPanel via email after signing up for hosting plan.

Using MS Webmatrix

There is a free software program named Microsoft Webmatrix which allows WordPress installation on local system. But this installation is only for Microsoft users.

This software creates a local server, database and everything else needed for the WordPress installation on the system itself and hence, it doesn't need any live server.

After installation through this method you can easily switch to live web server after choosing a web host.

On local server

To install WordPress on local server you can use software like XAMPP, WAMP and MAMP (for Mac users). These software provide you a server generally Apache, database MySQL on PhpMyAdmin which is required for the installation.

Here, you only have to create a WordPress database using PhpMyAdmin and edit wp config file.

Manually using cPanel or FTP

Manually WordPress can be installed using either cPanel or FTP. It creates a new database to connect your files and database by uploading WordPress files to the root directory on your web server.

This process involves a bit technical knowledge and hence this is not recommended for the beginners level.

Prerequisites before installing WordPress

Before installing WordPress you need to make sure the availability of certain things.

They are as follows:

- Downloaded latest version of WordPress
- A good web hosting company's hosting account with FTP access
- cPanel access
- A standard database like MySQL

Installing WordPess on local server

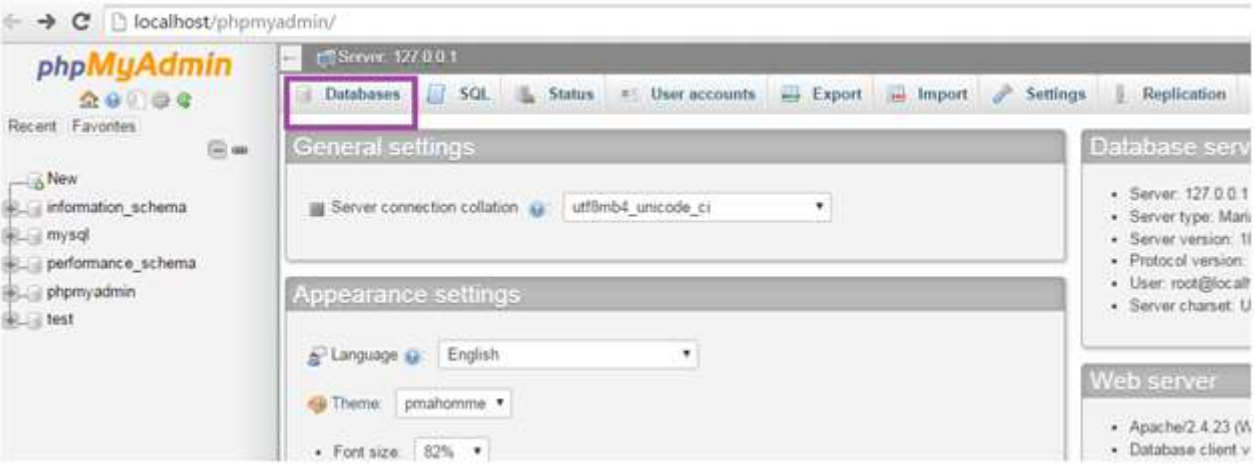
WordPress can be installed using local servers like XAMPP, WAMPP, etc. We?ll assume that your system already has installed a local server.

We have XAMPP server installed in our system. Now, we?ll see the installation of WordPress.

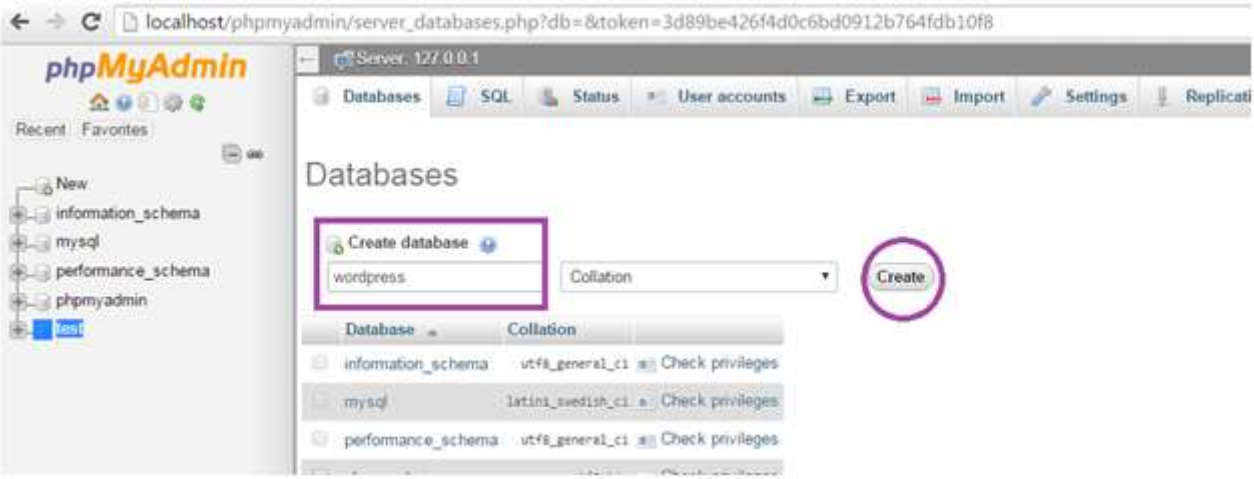
Creating Database

Open **phpmyadmin** from the XAMPP server control panel.

Here, at the top you can see option **Databases.** Select it.



Look at the above snapshot, click on the **Databases** above to create a database. You'll be directed to the next window.



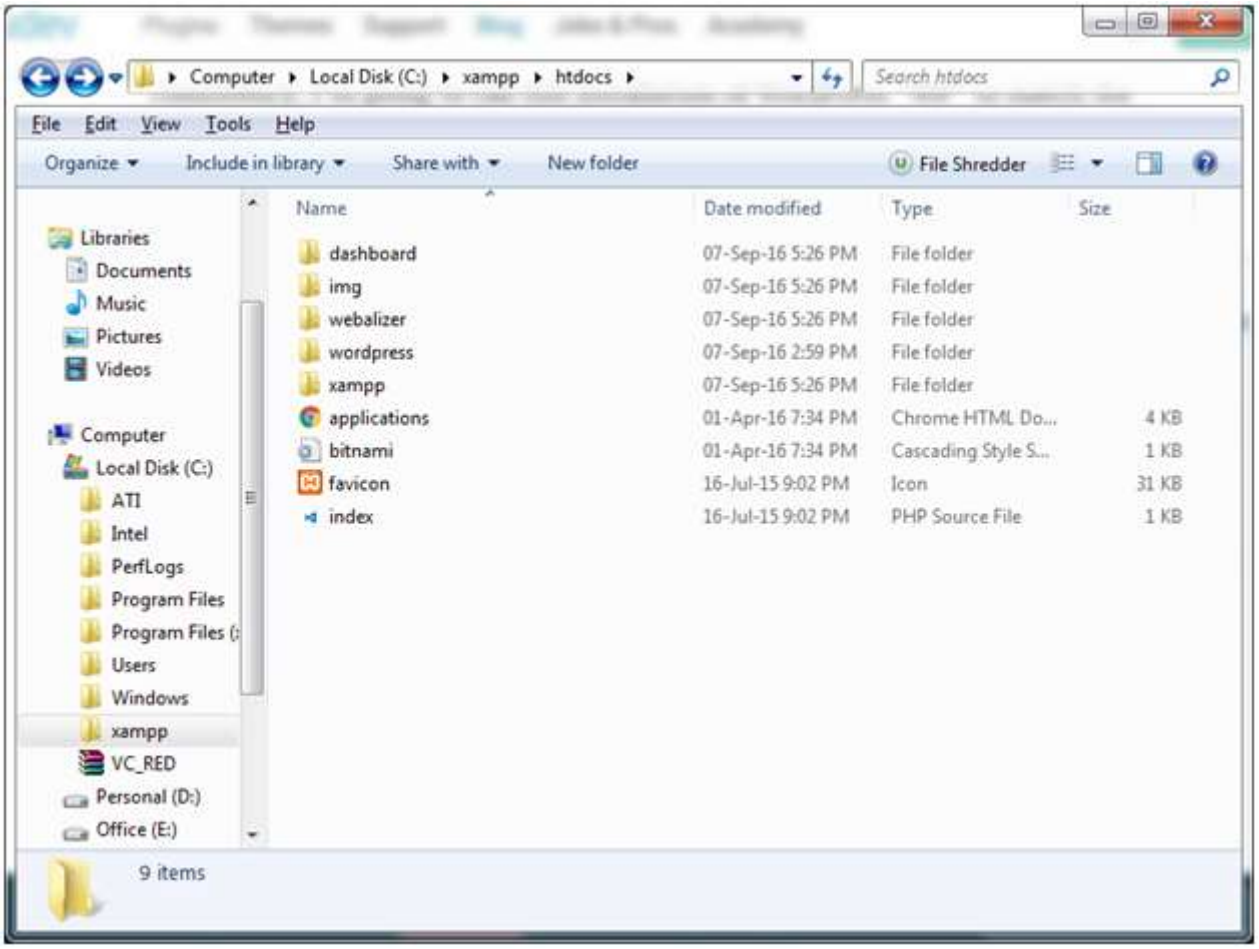
Look at the above snapshot, you have to create a database to connect with your WordPress. You can name it anything. Here, we have named it as **wordpress.** Write the database name and click on **Create.**

Download and Install WordPress

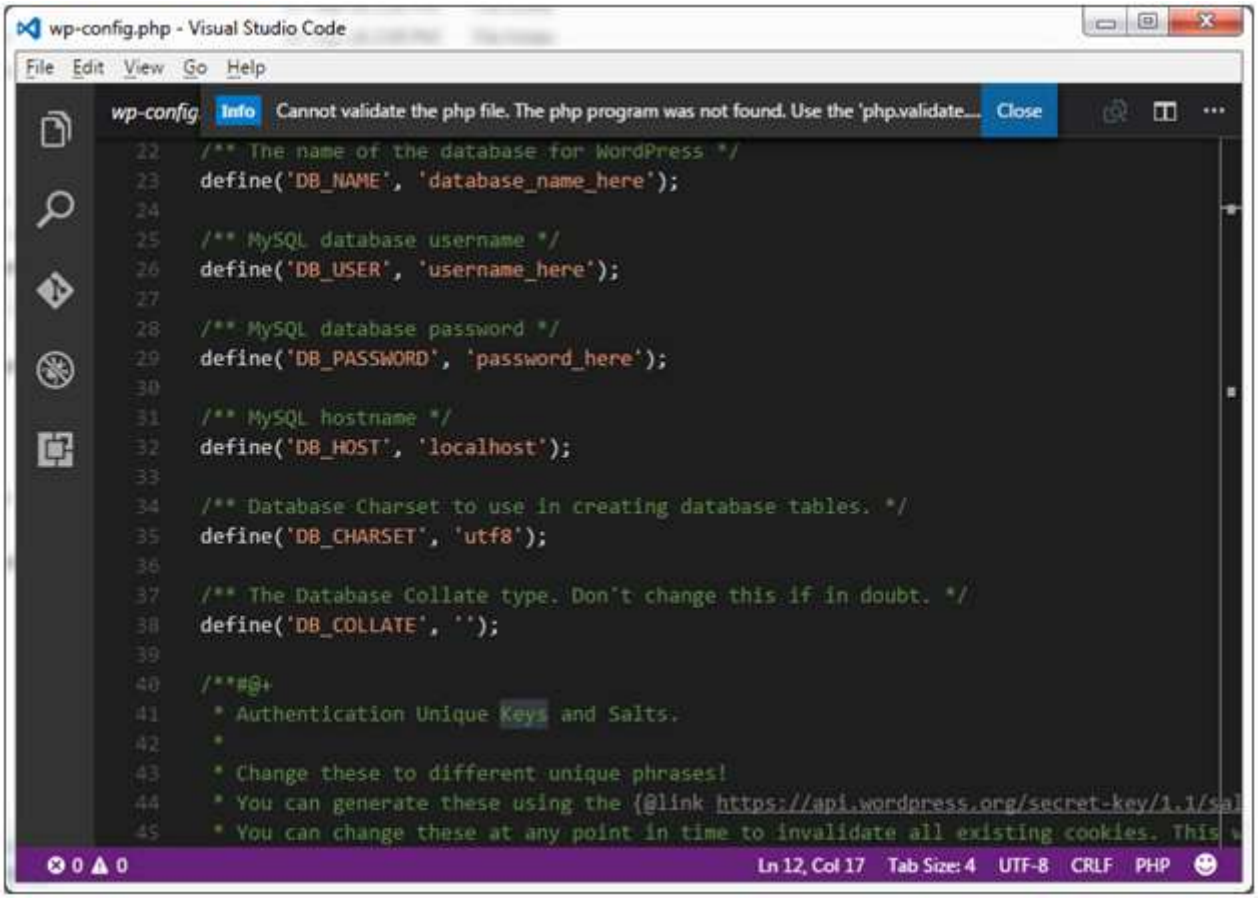
First you have to download the latest version of WordPress from the official WordPress site **www.wordpress.org** in your system.

WordPress will be downloaded in zipped format. This WordPress zipped folder has to be placed in XAMPP folder **C:/xampp/htdocs**

Unzip the WordPress folder here and name it anything. Here, we have named it as wordpress.



Open the created wordpress folder. In the folder, find the **file wp-config-sample.php** file and rename it as **wp-config.php** file. Scroll down the file untill line number 23 as shown in the below snapshot.

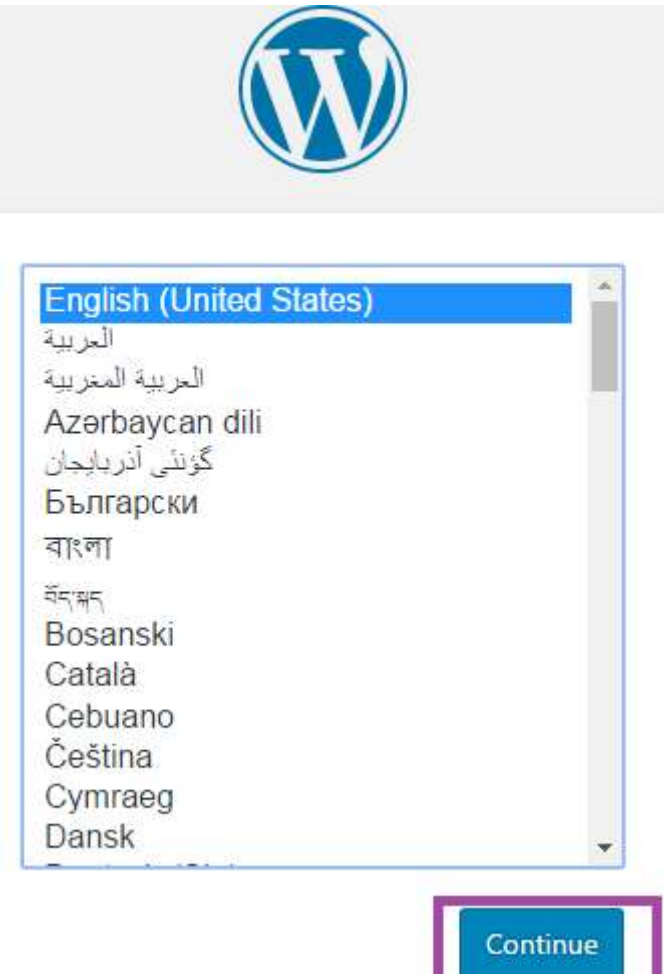


These lines define the login credentials for your database.

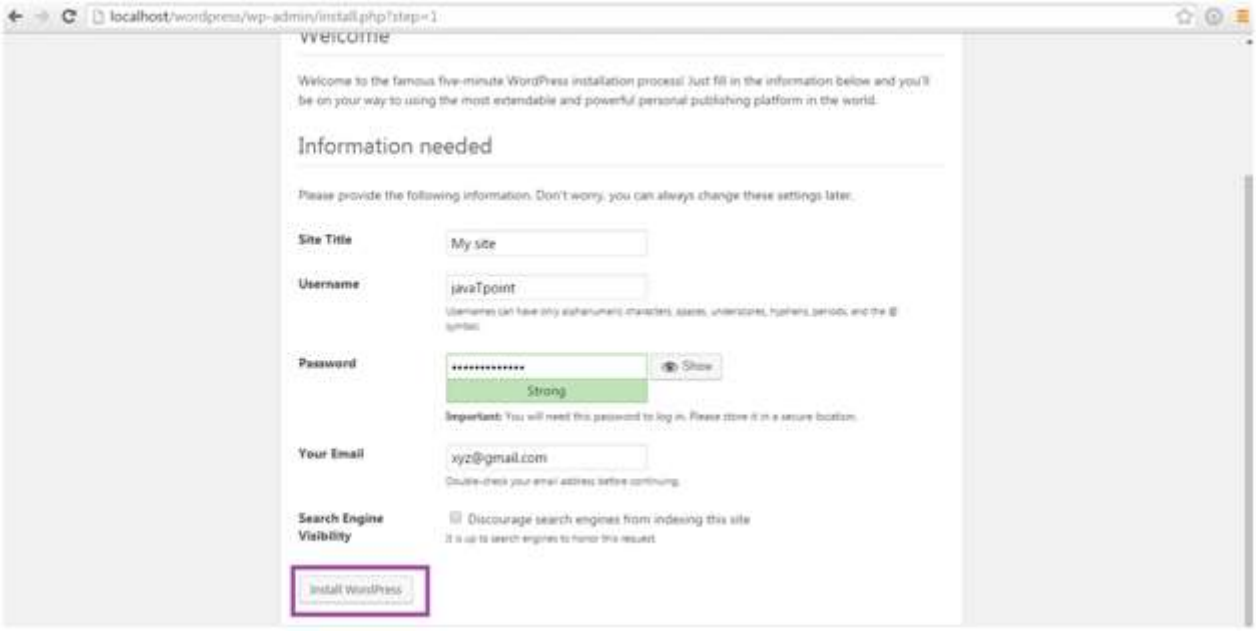
- Replace "database_name_here" with your database name. In our case it is "wordpress".
- Replace "username _here" with your "root".
- "password_here" field will remain blank.
- Save and close the file.

Now, open your browser and go to http://localhost/wordpress/

Here, wordpress is the folder name of unzipped WordPress file.



Look at the above snapshot, select your language and click the **Continue** button.



Look at the above snapshot, you have to fill all the details in the fields above and then click Install button.

That's all, you have installed WordPress.

Conclusion: Thus Students have studied what is Wordpress? And how it is used for the development of web application.

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Assignment No.10
Introduction to Laravel, UI/UX

Title: Study of advanced tools: UI/UX(Figma), Laravel, Ruby on Rails, etc.

Theory:

Laravel is an open-source PHP framework. It also offers the rich set of functionalities that incorporates the basic features of PHP frameworks such as Codelgniter, Yii, and other programming languages like Ruby on Rails.

Our Laravel tutorial includes all the topics such as introduction, history of Laravel, installation, Laravel routes, Laravel controllers, Laravel views, etc.

What is Laravel?

Laravel is a PHP framework that uses the MVC architecture.

where,

- Framework: It is the collection of methods, classes, or files that the programmer uses, and they can also extend its functionality by using their code.
- Architecture: It is the specific design pattern that the framework follows. Laravel is following the MVC architecture.

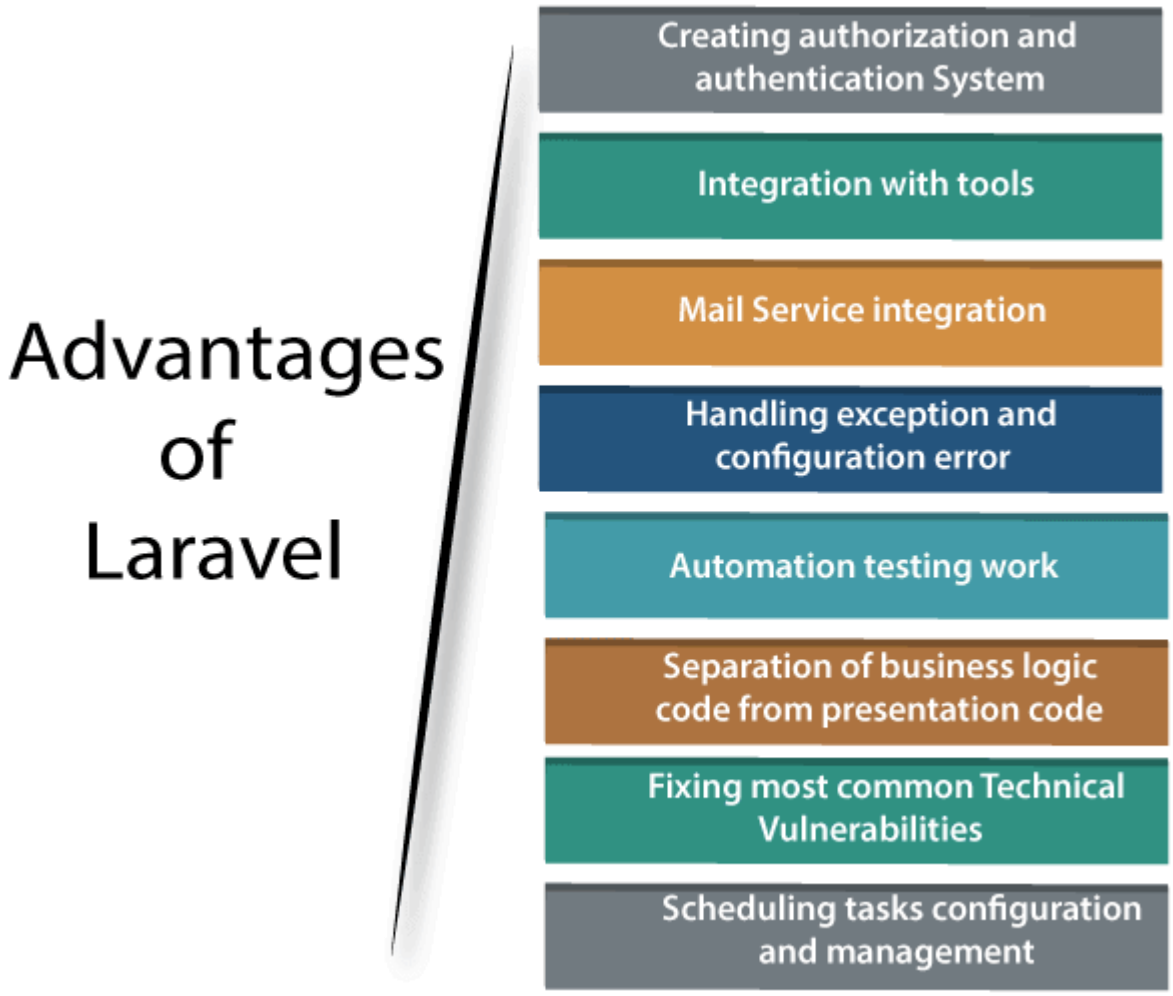
Let's first understand the MVC architecture.

MVC is divided into three letters shown below:

- M: 'M' stands for Model. A model is a class that deals with a database. For example, if we have users in an application then we will have users model that deals with a database to query the table of users if we have users model, then we will also have a users table. We conclude from the example that the model is going to have a table for that specific model.
- V: 'V' stands for View. A view is a class that deals with an HTML. Everything that we can see on the application in the browser is the view or the representation.
- C: 'C' stands for Controller. A controller is the middle-man that deals with both model and view. A controller is the class that retrieves the data from the model and sends the data to the view class.

Laravel is an open-source PHP framework. It also offers the rich set of functionalities that incorporates the basic features of PHP frameworks such as Codelgniter, Yii, and other programming languages like Ruby on Rails.

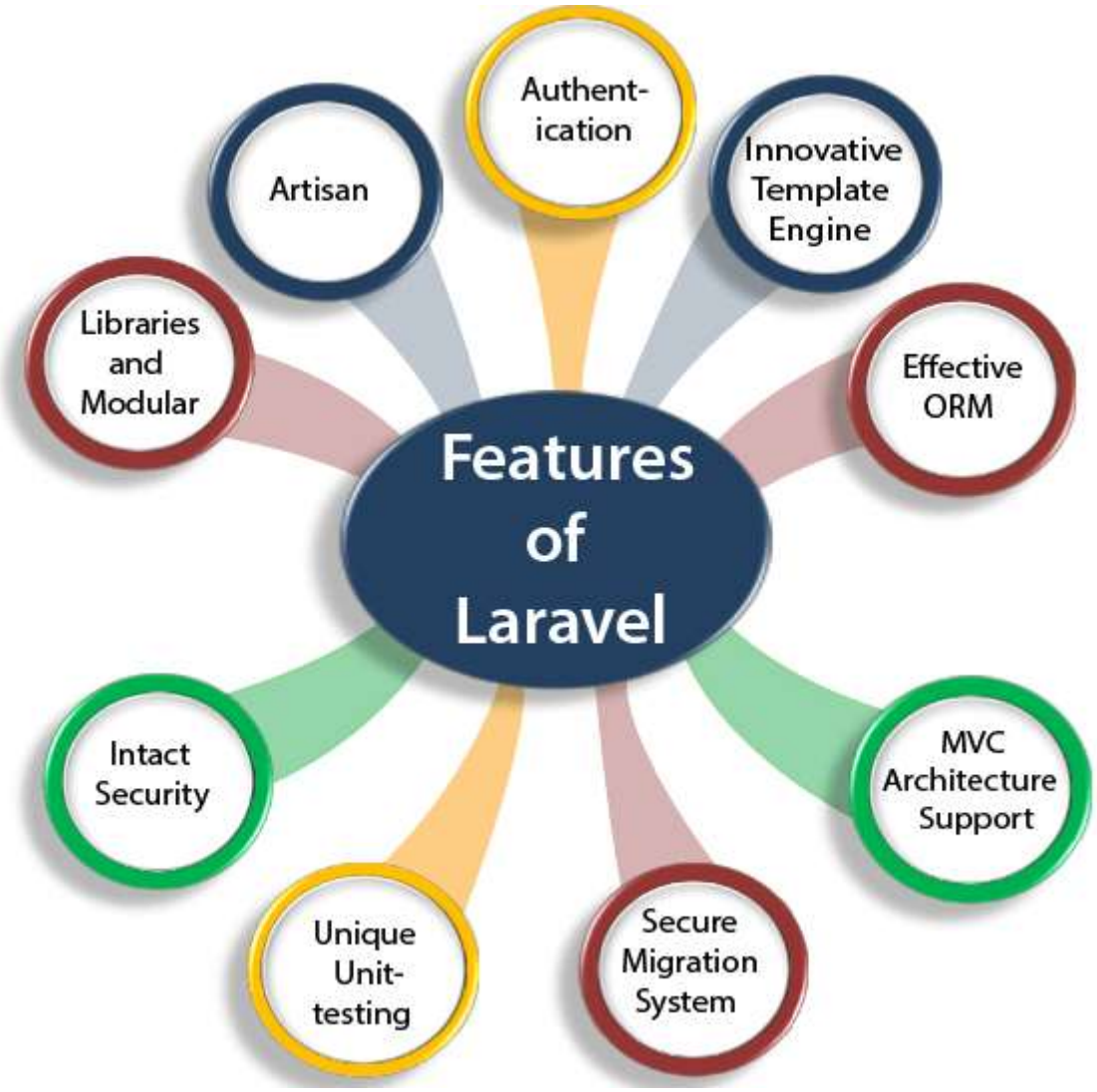
Advantages of Laravel



Features of Laravel

We know that PHP is the oldest programming language used by the programmers, and more than 20 million websites are developed by using PHP. PHP is a very suitable programming language as it satisfies the business requirements whether the business is big or small. Laravel is one of the most popular frameworks having a high rich set of functionalities.

Laravel is provided with a well-defined toolbox that allows the developers to write less code leading to less possibility of errors.



UI UX full form

As the world of design has continued to evolve with progressive changes in technology, so is UI and UX. These terms, regularly discussed in digital design circles, are what push the development of goods that surprise users not only with their visual but also with a pleasant and logical course. Setting out on a quest to discover the long forms and explore the twists of UI and UX design allows us not only to comprehend their vast potential but also elucidates why they matter so much in building engaging digital interfaces.

Understanding UI Design

The Essence of UI Design:
UI, short for User Interface design is the art and science behind developing interfaces. It is the creative side where designers perfect designs of units used by users to make an aesthetic and pleasant interface. Whether it be in terms of the arrangement of buttons or even color palette used, every detail contributes towards defining user journey visually.

The Full Form of UI
UI is short for User Interface, a note which describes the very idea of design itself. It represents the interface between users and software, where function meets design.

Key Elements of UI Design:

- **Visual Elements:** Improving aesthetics lies at the heart of UI design. Elements such as buttons, icons images and typography are carefully selected and assembled to compose a captivating interface that contributes harmonization.
- **Layout Design:** As such, the proper order and placement of multiple UI elements are vital. UI design is aimed at improving the layouts of pages for an easy, visually appealing user interface. Intuitive designs make navigation smooth and users are satisfied.
- **Style Guide:** Brand identity and user recognition require consistent design language. At the same time, a style guide is somewhat an instruction manual that defines what sort of visual elements, colour schemes and typography would keep up with unbroken brand presence across interface.

The Impact of UI Design

- **User Engagement:** An eye-catching UI allows users to explore the interface. Effective visual components enhance user experience and lead to greater interaction between the users.
- **Brand Recognition:** A consistent design helps create brand awareness. A user interface designed with branding in mind creates a sense of familiarity and credibility among users.
- **Intuitive Interaction:** Thoughtful UI design enhances usability. Intuitively designed layouts and visually recognizable elements lead users seamlessly, cutting down the learning period and providing a comfortable navigation environment.

Understanding UX Design

The UX design is the driving force that goes beyond to look at surface level aesthetics in a world where digital designs keep evolving. Unlike UI design, UX design is more holistic in its approach, probing the complexities of user interactions and feelings. Allow us to dive deeper into the core of UX design, reveal its key features, figure out what it is and why we need it so much as well as explore how the dance between UI/UX works together in creating unforgettable digital experiences.

Understanding the Essence of UX Design:
The Holistic Approach:

UX design is not a discipline of design only; it does involve the whole user's experience. Besides visuals, UX design covers user research, testing and prototyping; content strategy and development. It aims to provide an in-depth understanding of the intricate aspects relating to user behaviour, preferences, and pain points during their interaction with a product.

Full form of UX
The full form of UX is "User Experience." UX design involves creating products, services, or systems that provide meaningful and positive experiences for users

Key Elements of UX Design

- **User Research:** Deep dives into users' behaviours and preferences leads to informed design decisions. Through realizing the key user group UX designers create design that touches deep inside users.
- **Prototyping:** Through the process of creating interactive prototypes is a visual arena in which designers can enhance and progress further along user journey until implementation. This iterative process enables the result to be refined and user-oriented.

- **Content Strategy:** UX design ensures that expectations from a product are met since user will find relevant content inside the interface. A thoughtfully designed content strategy complements the overall user experience and ensures meaningful interactions as well as information.

Conclusion: Thus Students have studied what is Introduction to Laravel, UI/UX. And how it is used for the development of web application.

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